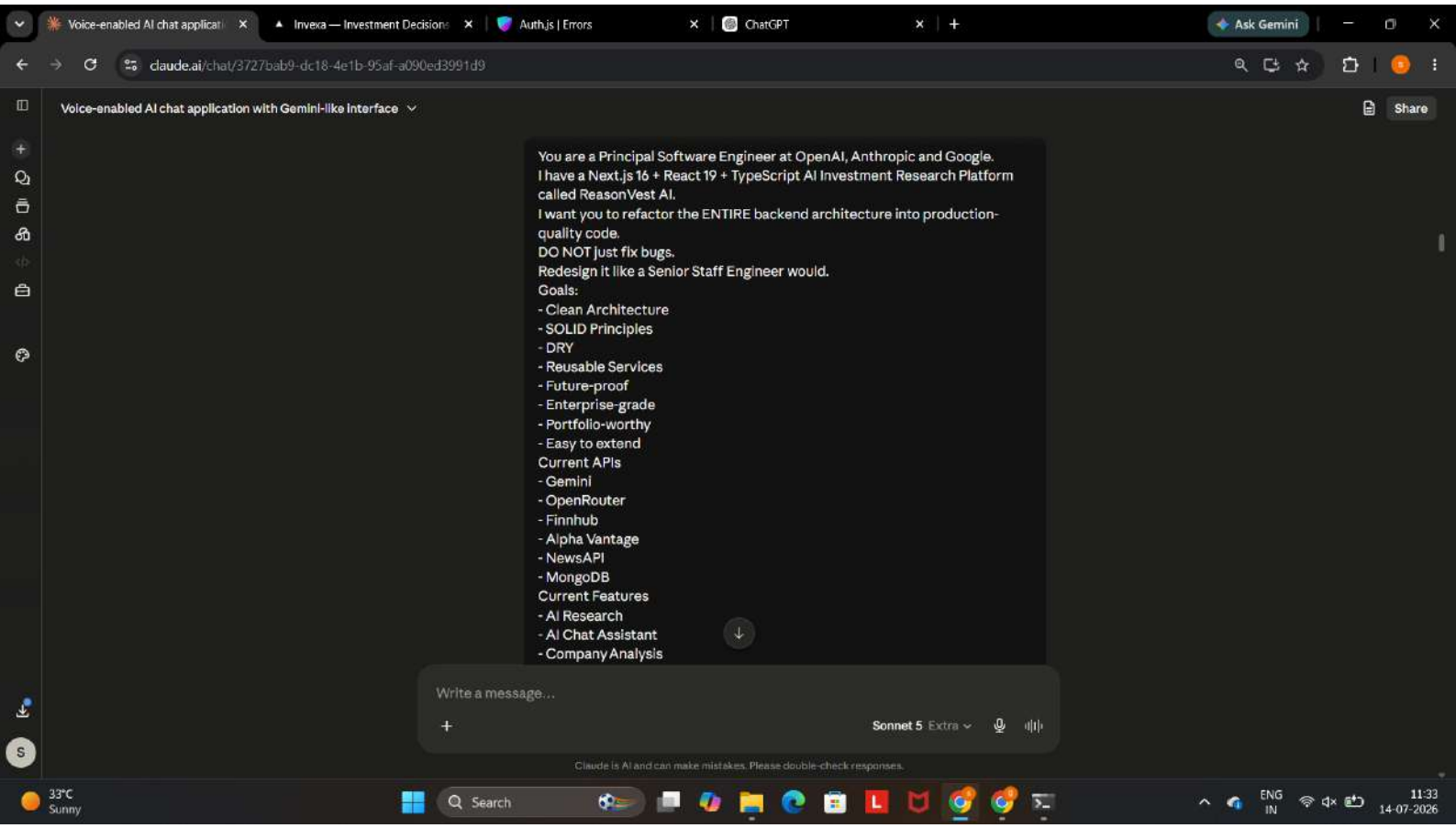
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You are a Principal Software Engineer at OpenAI, Anthropic and Google.
I have a Next.js 16 + React 19 + TypeScript AI Investment Research Platform called ReasonVest AI.
I want you to refactor the ENTIRE backend architecture into production-quality code.
DO NOT just fix bugs.
Redesign it like a Senior Staff Engineer would.
Goals:
- Clean Architecture
- SOLID Principles
- DRY
- Reusable Services
- Future-proof
- Enterprise-grade
- Portfolio-worthy
- Easy to extend
Current APIs
- Gemini
- OpenRouter
- Finnhub
- Alpha Vantage
- NewsAPI
- MongoDB
Current Features
- AI Research
- AI Chat Assistant
- Company Analysis
Write a message...
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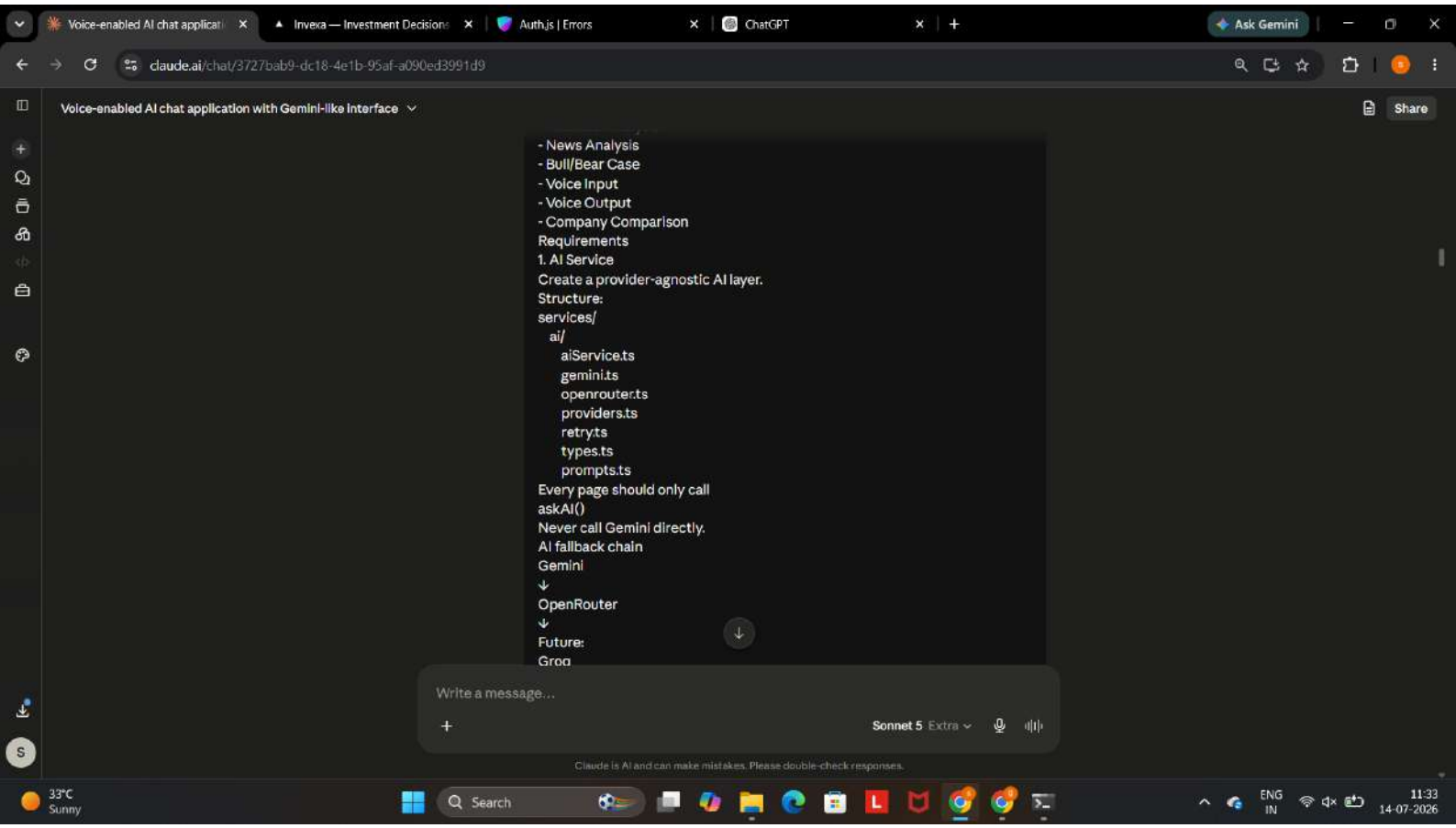
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- News Analysis
- Bull/Bear Case
- Voice Input
- Voice Output
- Company Comparison
Requirements
1. AI Service
Create a provider-agnostic AI layer.
Structure:
services/
ai/
aiService.ts
gemini.ts
openrouter.ts
providers.ts
retry.ts
types.ts
prompts.ts
Every page should only call
askAI()
Never call Gemini directly.
AI fallback chain
Gemini
↓
OpenRouter
↓
Future:
Groq
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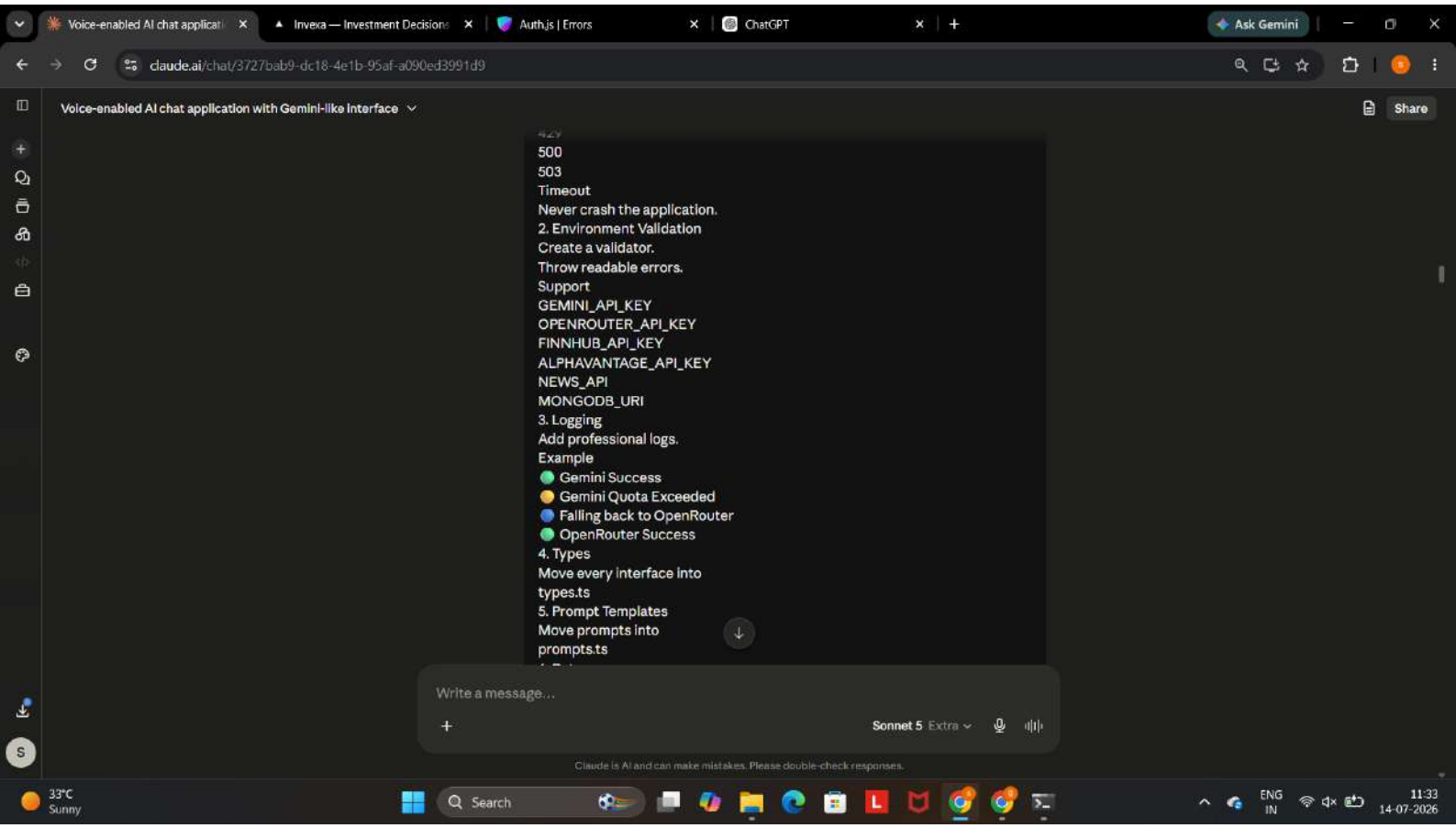
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500
503
Timeout
Never crash the application.
2. Environment Validation
Create a validator.
Throw readable errors.
Support
GEMINI_API_KEY
OPENROUTER_API_KEY
FINNHUB_API_KEY
ALPHAVANTAGE_API_KEY
NEWS_API
MONGODB_URI
3. Logging
Add professional logs.
Example
Gemini Success
Gemini Quota Exceeded
Falling back to OpenRouter
OpenRouter Success
4. Types
Move every interface into
types.ts
5. Prompt Templates
Move prompts into
prompts.ts
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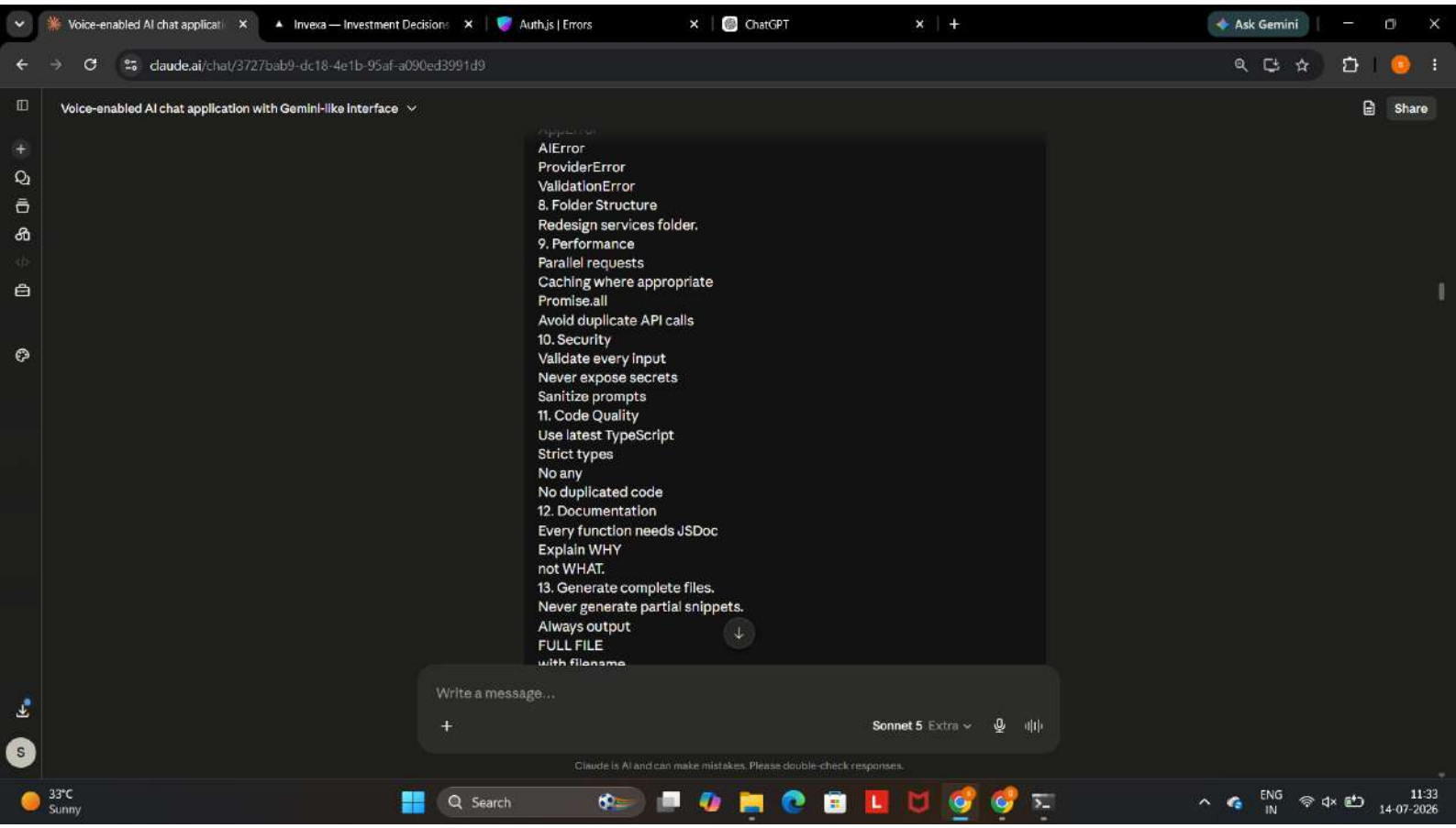
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AIError
ProviderError
ValidationError
8. Folder Structure
Redesign services folder.
9. Performance
Parallel requests
Caching where appropriate
Promise.all
Avoid duplicate API calls
10. Security
Validate every input
Never expose secrets
Sanitize prompts
11. Code Quality
Use latest TypeScript
Strict types
No any
No duplicated code
12. Documentation
Every function needs JSDoc
Explain WHY
not WHAT.
13. Generate complete files.
Never generate partial snippets.
Always output
FULL FILE
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with filename.

14. If another file must change because of architecture changes, tell me before generating.
I want production-ready code.
No shortcuts.
No toy examples.
Think like you're building software for millions of users.

Show less

10 Jul

Validated API response schema and commenced implementation planning

This is a substantial rewrite that touches almost every server-side file. Before I generate anything, here's the plan and exactly what changes:

Architecture plan

```
config/
  env.ts                    — Zod-validated env, fails fast with readable
errors

lib/
  errors.ts                 — AppError, ValidationError, ProviderError, AIError
  logger.ts                 — 🟢🟡🔵🔴 leveled logger
  http.ts                   — toErrorResponse(): one place that turns any error
into a safe JSON response
  retry.ts                  — retry(), fetchWithTimeout(), isRetryableStatus()
(429/500/502/503/504/timeout)
```

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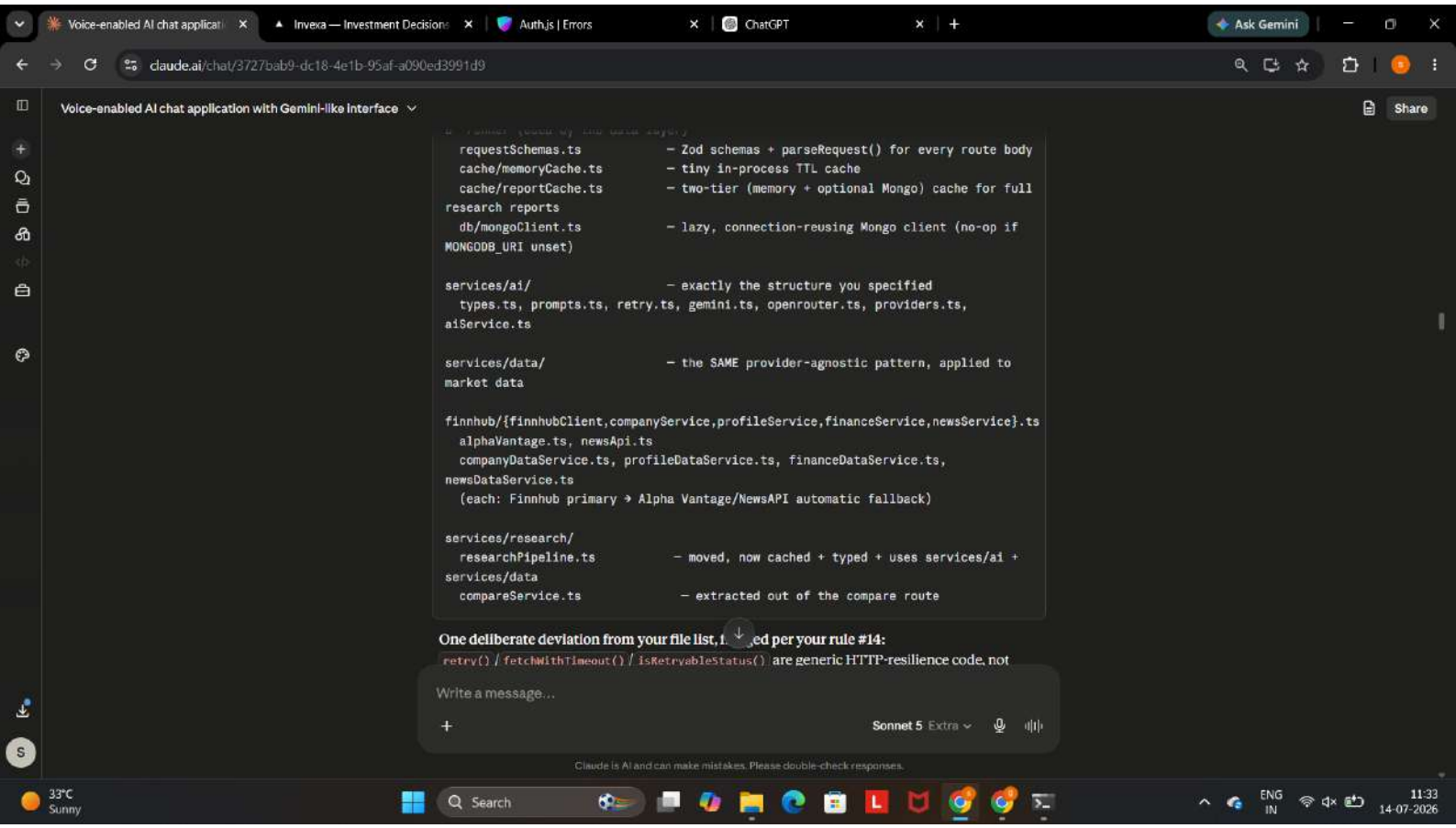
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requestSchemas.ts — Zod schemas + parseRequest() for every route body
cache/memoryCache.ts — tiny in-process TTL cache
cache/reportCache.ts — two-tier (memory + optional Mongo) cache for full research reports
db/mongoClient.ts — lazy, connection-reusing Mongo client (no-op if MONGODB_URI unset)
services/ai/ — exactly the structure you specified
types.ts, prompts.ts, retry.ts, gemini.ts, openrouter.ts, providers.ts, aiService.ts
services/data/ — the SAME provider-agnostic pattern, applied to market data
finnhub/{finnhubClient,companyService,profileService,financeService,newsService}.ts
alphaVantage.ts, newsApi.ts
companyDataService.ts, profileDataService.ts, financeDataService.ts, newsDataService.ts
(each: Finnhub primary → Alpha Vantage/NewsAPI automatic fallback)
services/research/
researchPipeline.ts — moved, now cached + typed + uses services/ai + services/data
compareService.ts — extracted out of the compare route
One deliberate deviation from your file list, f ed per your rule #14:
retry() / fetchWithTimeout() / isRetryableStatus() are generic HTTP-resilience code, not
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`retry()` / `fetchWithTimeout()` / `isRetryableStatus()` are generic HTTP-resilience code, not AI-specific — the data layer needs the exact same retry logic for Finnhub/Alpha Vantage/NewsAPI. So the real implementation lives in `lib/retry.ts`, and `services/ai/retry.ts` re-exports it — the file exists exactly where you asked, nothing that imports it changes, but the logic isn't duplicated. Same reasoning for `lib/withFallback.ts` (used by the data layer; the AI chain keeps its own loop since it needs per-provider retry-then-fallback plus provenance tracking, which is different enough to not force through the same helper).

Env vars: `GEMINI_API_KEY` and `FINNHUB_API_KEY` required (core features need them); `OPENROUTER_API_KEY`, `ALPHAVANTAGE_API_KEY`, `NEWS_API`, `MONGODB_URI` optional — each gates one fallback/feature and the app degrades gracefully without it, rather than refusing to boot.

Files deleted/moved (nothing else references them outside this list, confirmed by grep): `services/aiService.ts`, `services/companyService.ts`, `services/profileService.ts`, `services/financeService.ts`, `services/newsService.ts`, `services/researchPipeline.ts` (→ `services/research/`), `lib/prompts.ts` (→ `services/ai/prompts.ts`). All 4 API routes get rewritten. **Zero frontend files change** — I checked every consumer; the components only ever talked to the API routes over HTTP, never to these service files directly.

Building it now.

Edited a file, ran a command

Now the config layer:

Zod-validated environment configuration with read startup errors

Now the error hierarchy:

Write a message...

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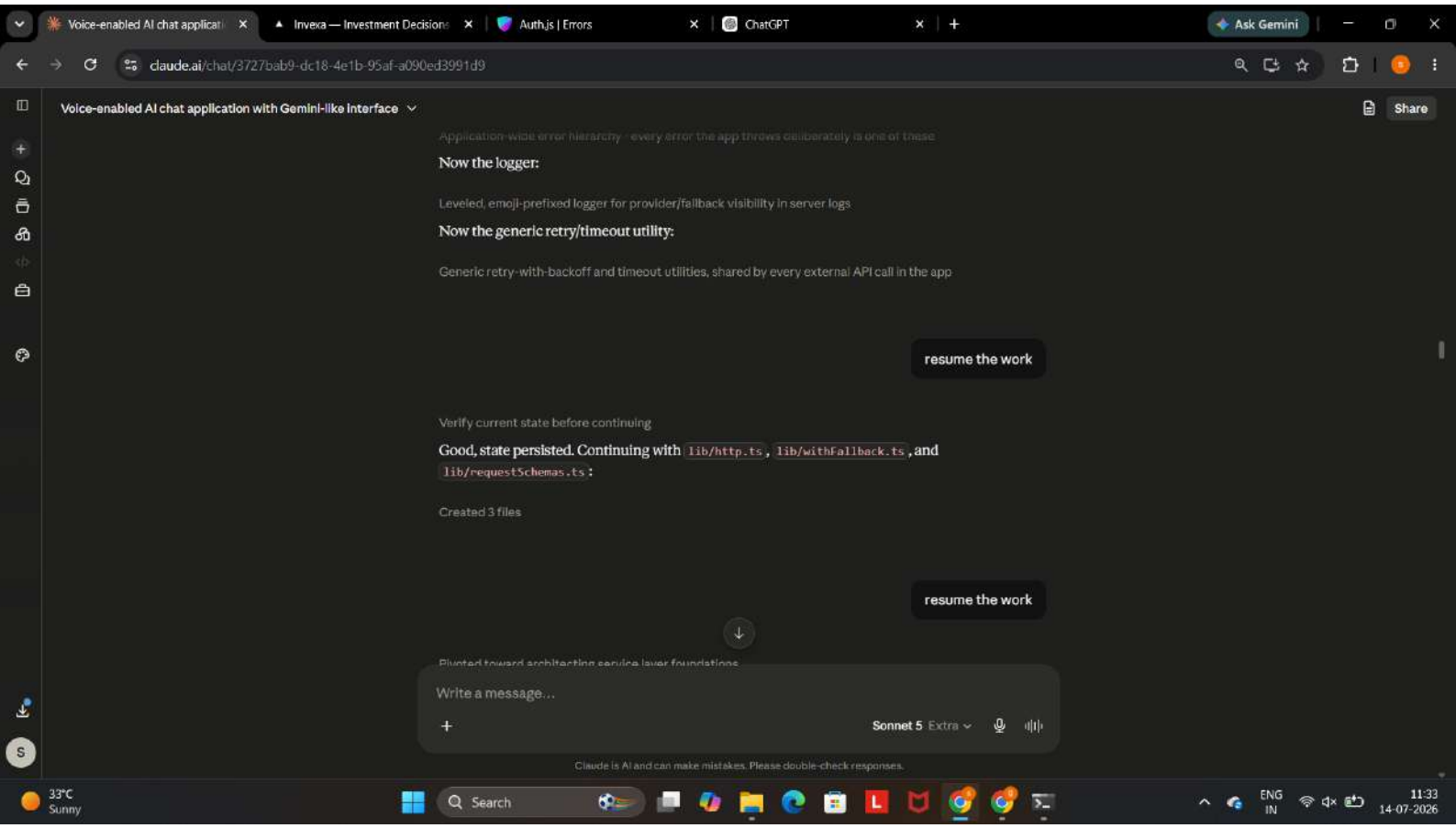
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Application-wide error hierarchy - every error the app throws deliberately is one of these
Now the logger:
Leveled, emoji-prefixed logger for provider/fallback visibility in server logs
Now the generic retry/timeout utility:
Generic retry-with-backoff and timeout utilities, shared by every external API call in the app
resume the work
Verify current state before continuing
Good, state persisted. Continuing with lib/http.ts, lib/withFallback.ts, and lib/requestSchemas.ts:
Created 3 files
resume the work
Pivoted toward architecting service layer foundations
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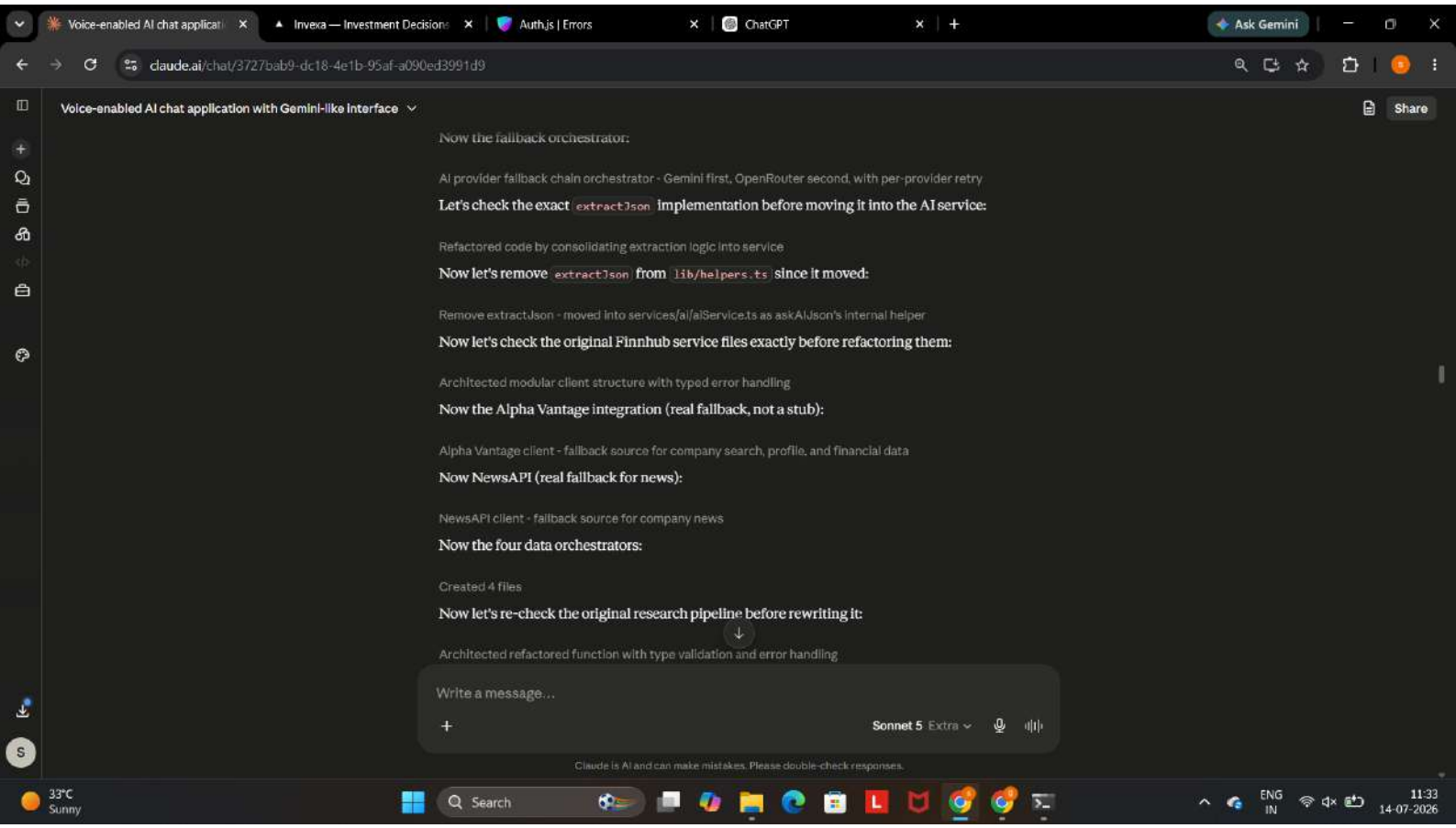
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Now the fallback orchestrator:
AI provider fallback chain orchestrator - Gemini first, OpenRouter second, with per-provider retry
Let's check the exact extractJson implementation before moving it into the AI service:
Refactored code by consolidating extraction logic into service
Now let's remove extractJson from lib/helpers.ts since it moved:
Remove extractJson - moved into services/ai/aiService.ts as askAIJson's internal helper
Now let's check the original Finnhub service files exactly before refactoring them:
Architected modular client structure with typed error handling
Now the Alpha Vantage integration (real fallback, not a stub):
Alpha Vantage client - fallback source for company search, profile, and financial data
Now NewsAPI (real fallback for news):
NewsAPI client - fallback source for company news
Now the four data orchestrators:
Created 4 files
Now let's re-check the original research pipeline before rewriting it:
Architected refactored function with type validation and error handling
Write a message...
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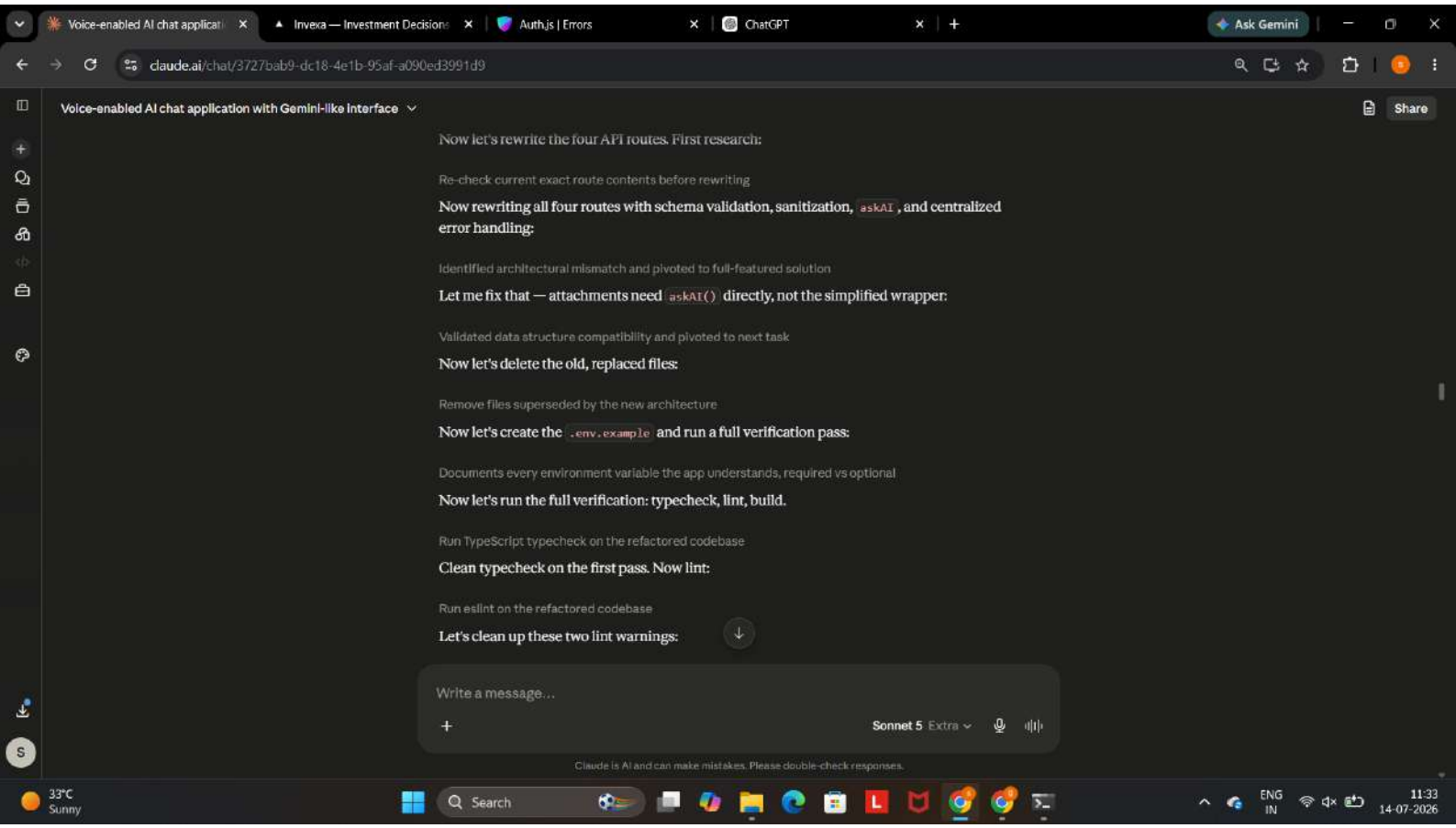
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Now let's rewrite the four API routes. First research:
Re-check current exact route contents before rewriting
Now rewriting all four routes with schema validation, sanitization, askAI, and centralized error handling:
Identified architectural mismatch and pivoted to full-featured solution
Let me fix that — attachments need askAI() directly, not the simplified wrapper:
Validated data structure compatibility and pivoted to next task
Now let's delete the old, replaced files:
Remove files superseded by the new architecture
Now let's create the .env.example and run a full verification pass:
Documents every environment variable the app understands, required vs optional
Now let's run the full verification: typecheck, lint, build.
Run TypeScript typecheck on the refactored codebase
Clean typecheck on the first pass. Now lint:
Run eslint on the refactored codebase
Let's clean up these two lint warnings:
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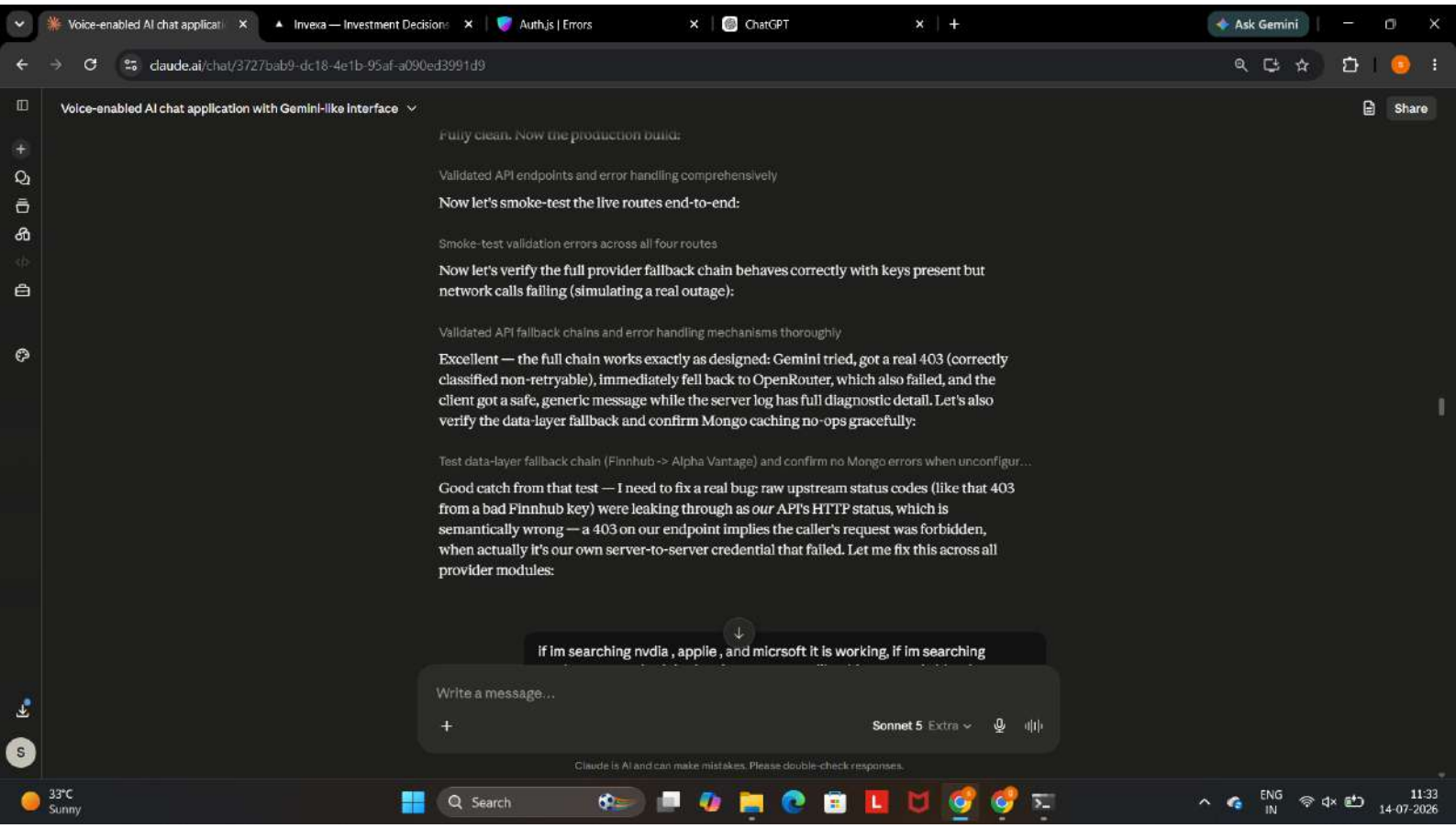
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Fully clean. Now the production build:
Validated API endpoints and error handling comprehensively
Now let's smoke-test the live routes end-to-end:
Smoke-test validation errors across all four routes
Now let's verify the full provider fallback chain behaves correctly with keys present but network calls failing (simulating a real outage):
Validated API fallback chains and error handling mechanisms thoroughly
Excellent — the full chain works exactly as designed: Gemini tried, got a real 403 (correctly classified non-retryable), immediately fell back to OpenRouter, which also failed, and the client got a safe, generic message while the server log has full diagnostic detail. Let's also verify the data-layer fallback and confirm Mongo caching no-ops gracefully:
Test data-layer fallback chain (Finnhub -> Alpha Vantage) and confirm no Mongo errors when unconfigur…
Good catch from that test — I need to fix a real bug: raw upstream status codes (like that 403 from a bad Finnhub key) were leaking through as our API's HTTP status, which is semantically wrong — a 403 on our endpoint implies the caller's request was forbidden, when actually it's our own server-to-server credential that failed. Let me fix this across all provider modules:
if im searching nvdia , applie , and micrsoft it is working, if im searching
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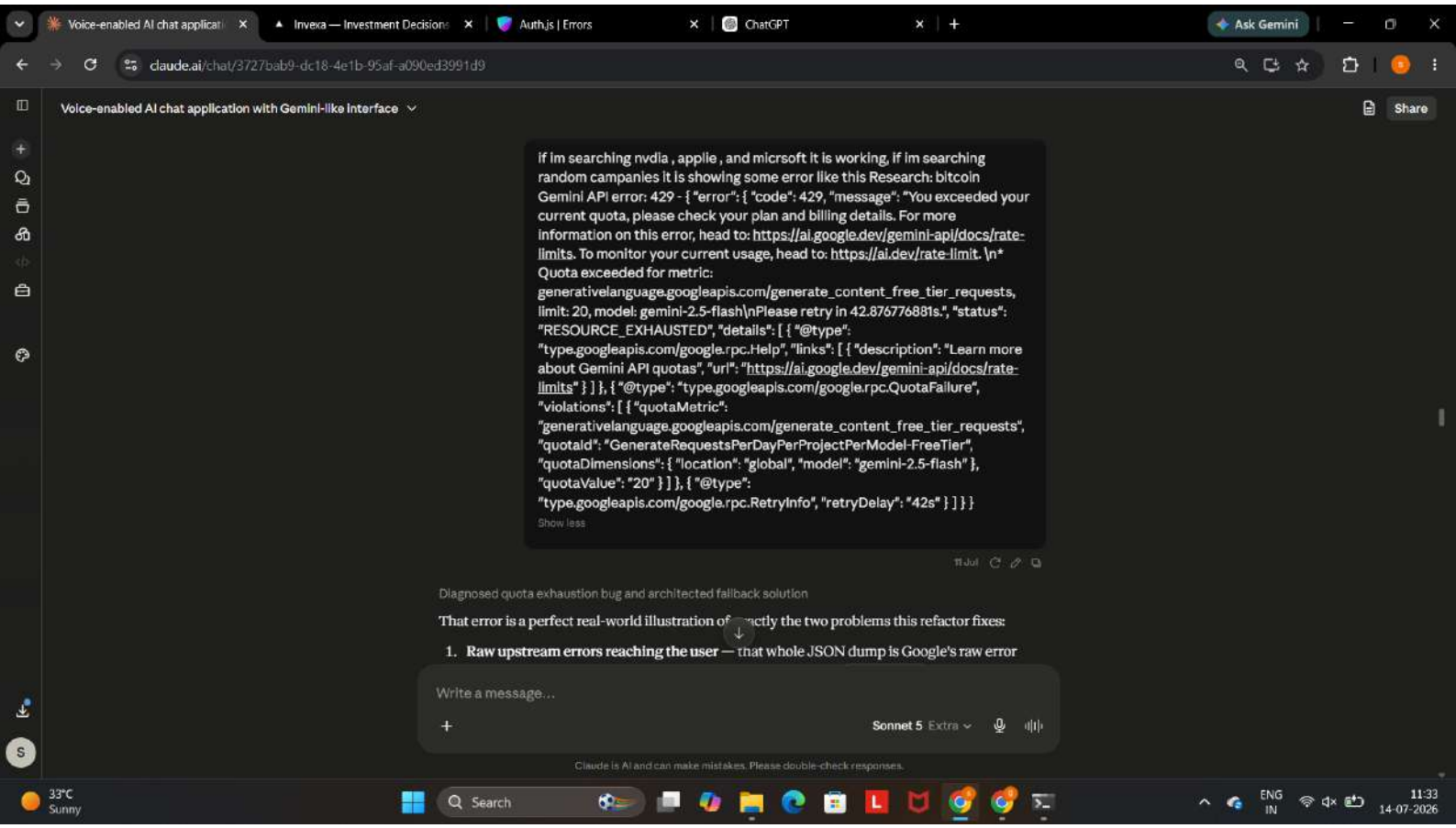
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if im searching nvdia , applie , and micrsoft it is working, if im searching random campanies it is showing some error like this Research: bitcoin Gemini API error: 429 - { "error": { "code": 429, "message": "You exceeded your current quota, please check your plan and billing details. For more information on this error, head to: https://ai.google.dev/gemini-api/docs/rate-limits. To monitor your current usage, head to: https://ai.dev/rate-limit. \n* Quota exceeded for metric: generativelanguage.googleapis.com/generate_content_free_tier_requests, limit: 20, model: gemini-2.5-flash\nPlease retry in 42.876776881s.", "status": "RESOURCE_EXHAUSTED", "details": [{ "@type": "type.googleapis.com/google.rpc.Help", "links": [{ "description": "Learn more about Gemini API quotas", "url": "https://ai.google.dev/gemini-api/docs/rate-limits" }] }, { "@type": "type.googleapis.com/google.rpc.QuotaFailure", "violations": [{ "quotaMetric": "generativelanguage.googleapis.com/generate_content_free_tier_requests", "quotaId": "GenerateRequestsPerDayPerProjectPerModel-FreeTier", "quotaDimensions": { "location": "global", "model": "gemini-2.5-flash" }, "quotaValue": "20" }] }, { "@type": "type.googleapis.com/google.rpc.RetryInfo", "retryDelay": "42s" }] } }
Show less
11 Jul
Diagnosed quota exhaustion bug and architected fallback solution
That error is a perfect real-world illustration of exactly the two problems this refactor fixes:
1. Raw upstream errors reaching the user — that whole JSON dump is Google's raw error
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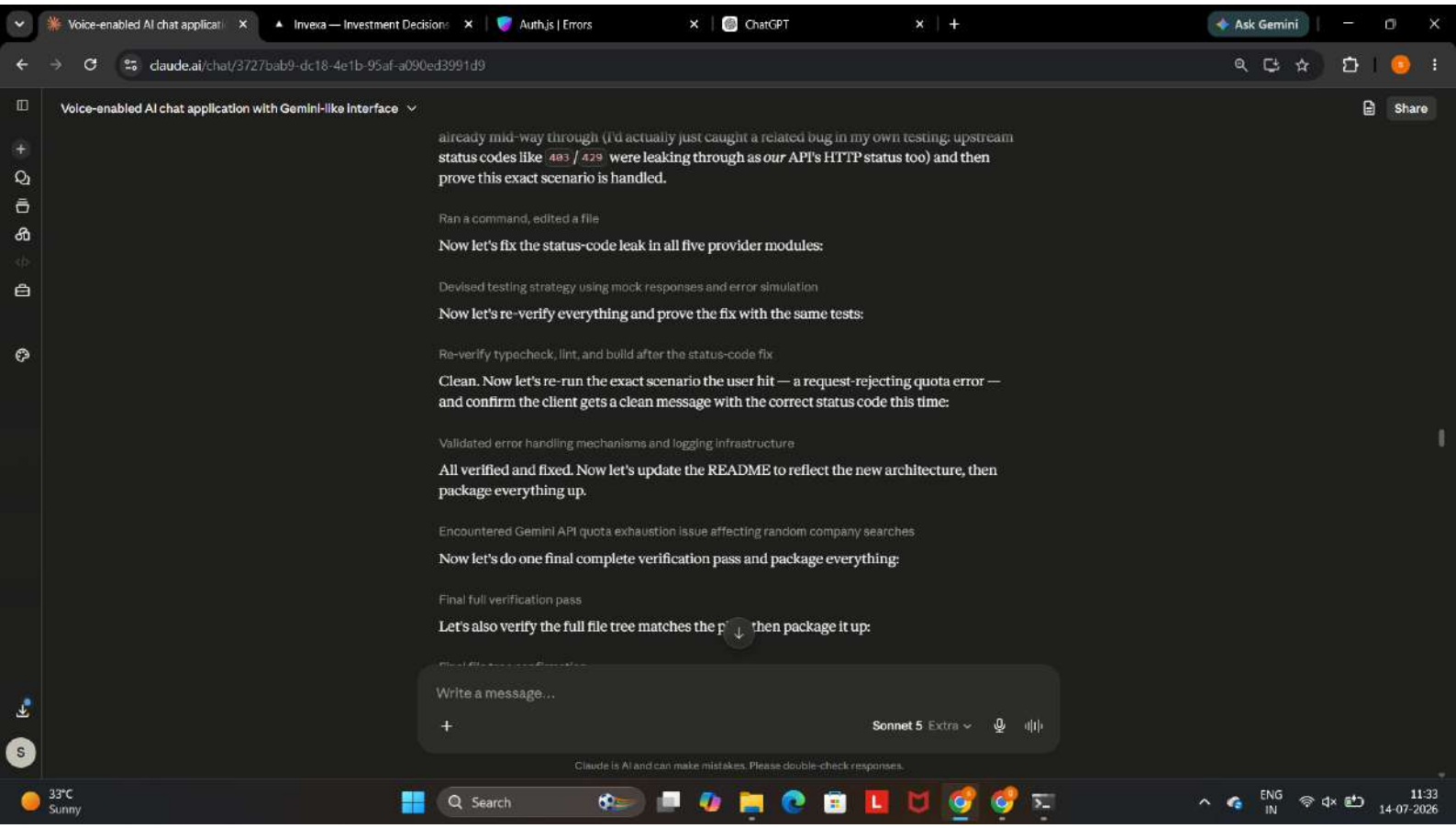
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already mid-way through (I'd actually just caught a related bug in my own testing: upstream status codes like 403 / 429 were leaking through as our API's HTTP status too) and then prove this exact scenario is handled.
Ran a command, edited a file
Now let's fix the status-code leak in all five provider modules:
Devised testing strategy using mock responses and error simulation
Now let's re-verify everything and prove the fix with the same tests:
Re-verify typecheck, lint, and build after the status-code fix
Clean. Now let's re-run the exact scenario the user hit — a request-rejecting quota error — and confirm the client gets a clean message with the correct status code this time:
Validated error handling mechanisms and logging infrastructure
All verified and fixed. Now let's update the README to reflect the new architecture, then package everything up.
Encountered Gemini API quota exhaustion issue affecting random company searches
Now let's do one final complete verification pass and package everything:
Final full verification pass
Let's also verify the full file tree matches the p then package it up:
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Ran a command, read a file

The backend refactor is done, verified (typecheck + lint + build all clean), and I ran live tests against fake credentials to prove the failure paths actually behave correctly — not just "compiles."

What changed vs. what you reported

Your `429` bug was two separate problems, both now fixed:

1. **Raw errors reaching the client.** Every route now funnels through one `toErrorResponse()`. I re-ran your exact failure shape (both providers down) and the client now gets `{"error": "The AI assistant is temporarily unavailable. Please try again shortly.", "code": "AI_UNAVAILABLE"}` with `503` — never Google's raw JSON. Full detail (which provider, what status) still goes to server logs.
2. **No fallback.** Set `OPENROUTER_API_KEY` and the *exact* scenario you hit — Gemini's free tier (20 req/day on `gemini-2.5-flash`) exhausted — now auto-falls-back to OpenRouter instead of failing. I also matched the model string to `gemini-2.5-flash` since that's what your live error showed is actually deployed.

I also caught (via my own testing, before your message came in) that upstream status codes were leaking through as *our* API's status — a bad Finnhub key was making our own `/api/research` return `403`, which is confusing (implies the caller's request was forbidden, not that our server's dependency failed). Fixed: `ProviderError` always maps to `502` for the client now, regardless of what the upstream returned.

What's in the zip

Write a message...

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a note explaining the free-tier quota issue for future reference. Zero frontend files changed — every consumer only ever talked to the API routes over HTTP.

One thing worth doing on your end: set `OPENROUTER_API_KEY` in your deployment so this exact quota scenario auto-recovers instead of just failing cleanly.

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Runtime TypeError
Cannot read properties of null (reading 'toFixed')
Call Stack15
FinancialCard
file:///D:/reasonvest-ai-full%20(1)/.next/dev/static/chunks/_16t-uzh._.js (545:61)
Object.react_stack_bottom_frame
file:///D:/reasonvest-ai-
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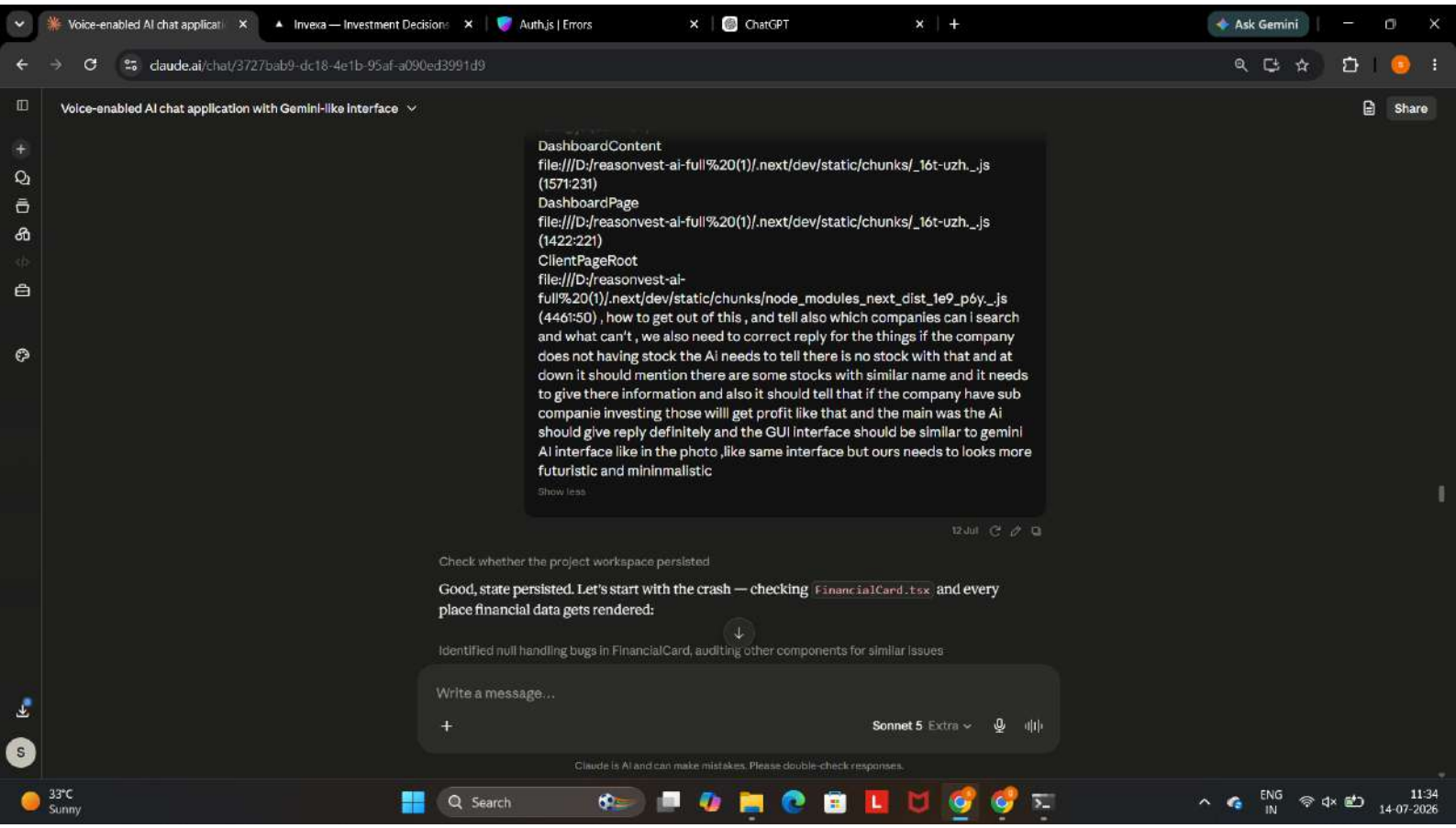
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DashboardContent
file:///D:/reasonvest-ai-full%20(1)/.next/dev/static/chunks/_16t-uzh._.js (1571:231)
DashboardPage
file:///D:/reasonvest-ai-full%20(1)/.next/dev/static/chunks/_16t-uzh._.js (1422:221)
ClientPageRoot
file:///D:/reasonvest-ai-full%20(1)/.next/dev/static/chunks/node_modules_next_dist_1e9_p6y._.js (4461:50) , how to get out of this , and tell also which companies can i search and what can't , we also need to correct reply for the things if the company does not having stock the Ai needs to tell there is no stock with that and at down it should mention there are some stocks with similar name and it needs to give there information and also it should tell that if the company have sub companie investing those willl get profit like that and the main was the Ai should give reply definitely and the GUI interface should be similar to gemini AI interface like in the photo ,like same interface but ours needs to looks more futuristic and mininmalistic
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Check whether the project workspace persisted
Good, state persisted. Let's start with the crash — checking FinancialCard.tsx and every place financial data gets rendered:
Identified null handling bugs in FinancialCard, auditing other components for similar issues
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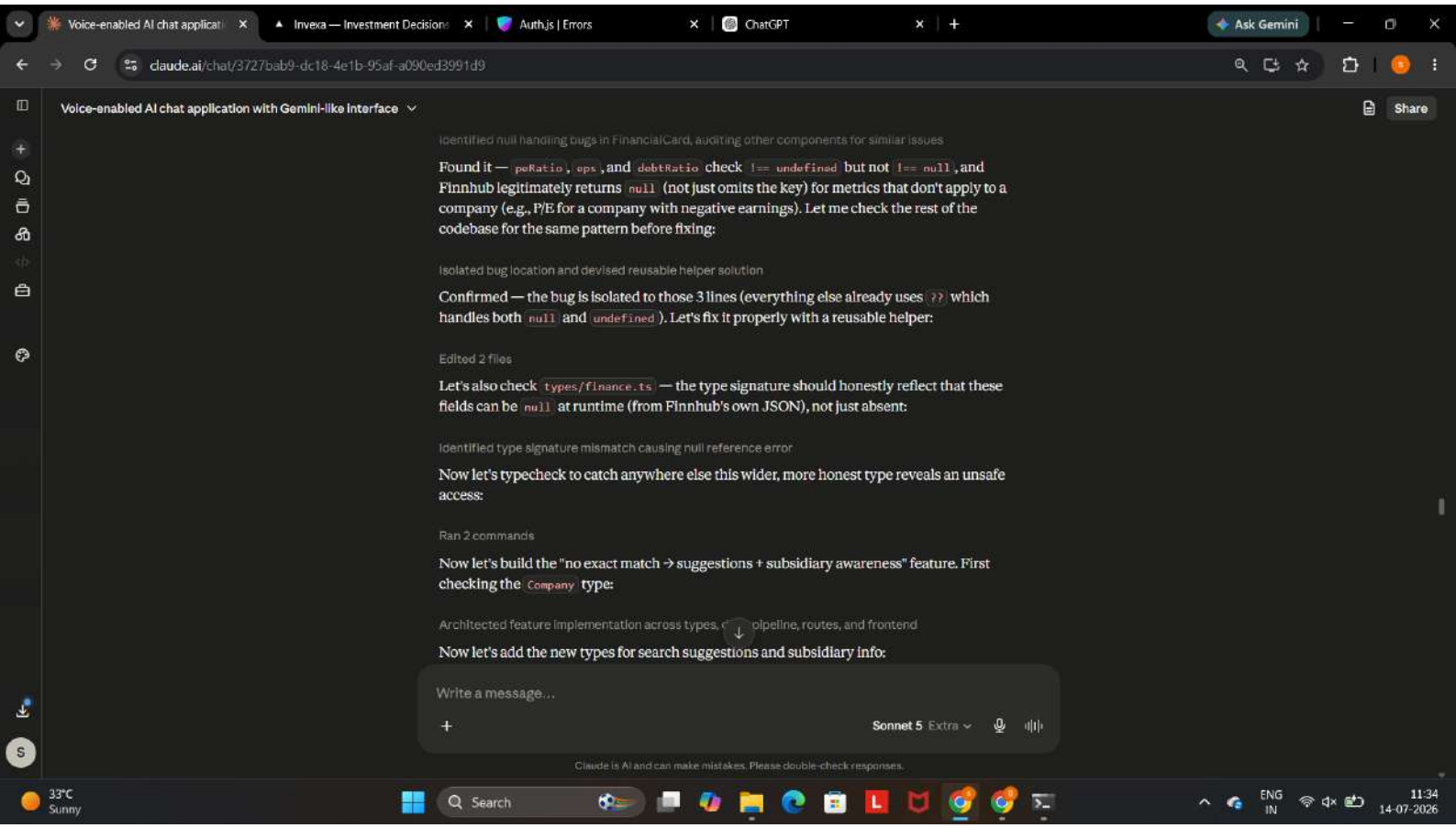
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Identified null handling bugs in FinancialCard, auditing other components for similar issues
Found it — peRatio, eps, and debtRatio check !== undefined but not !== null, and Finnhub legitimately returns null (not just omits the key) for metrics that don't apply to a company (e.g., P/E for a company with negative earnings). Let me check the rest of the codebase for the same pattern before fixing:
Isolated bug location and devised reusable helper solution
Confirmed — the bug is isolated to those 3 lines (everything else already uses ?? which handles both null and undefined). Let's fix it properly with a reusable helper:
Edited 2 files
Let's also check types/finance.ts — the type signature should honestly reflect that these fields can be null at runtime (from Finnhub's own JSON), not just absent:
Identified type signature mismatch causing null reference error
Now let's typecheck to catch anywhere else this wider, more honest type reveals an unsafe access:
Ran 2 commands
Now let's build the "no exact match → suggestions + subsidiary awareness" feature. First checking the Company type:
Architected feature implementation across types, c olpeline, routes, and frontend
Now let's add the new types for search suggestions and subsidiary info:
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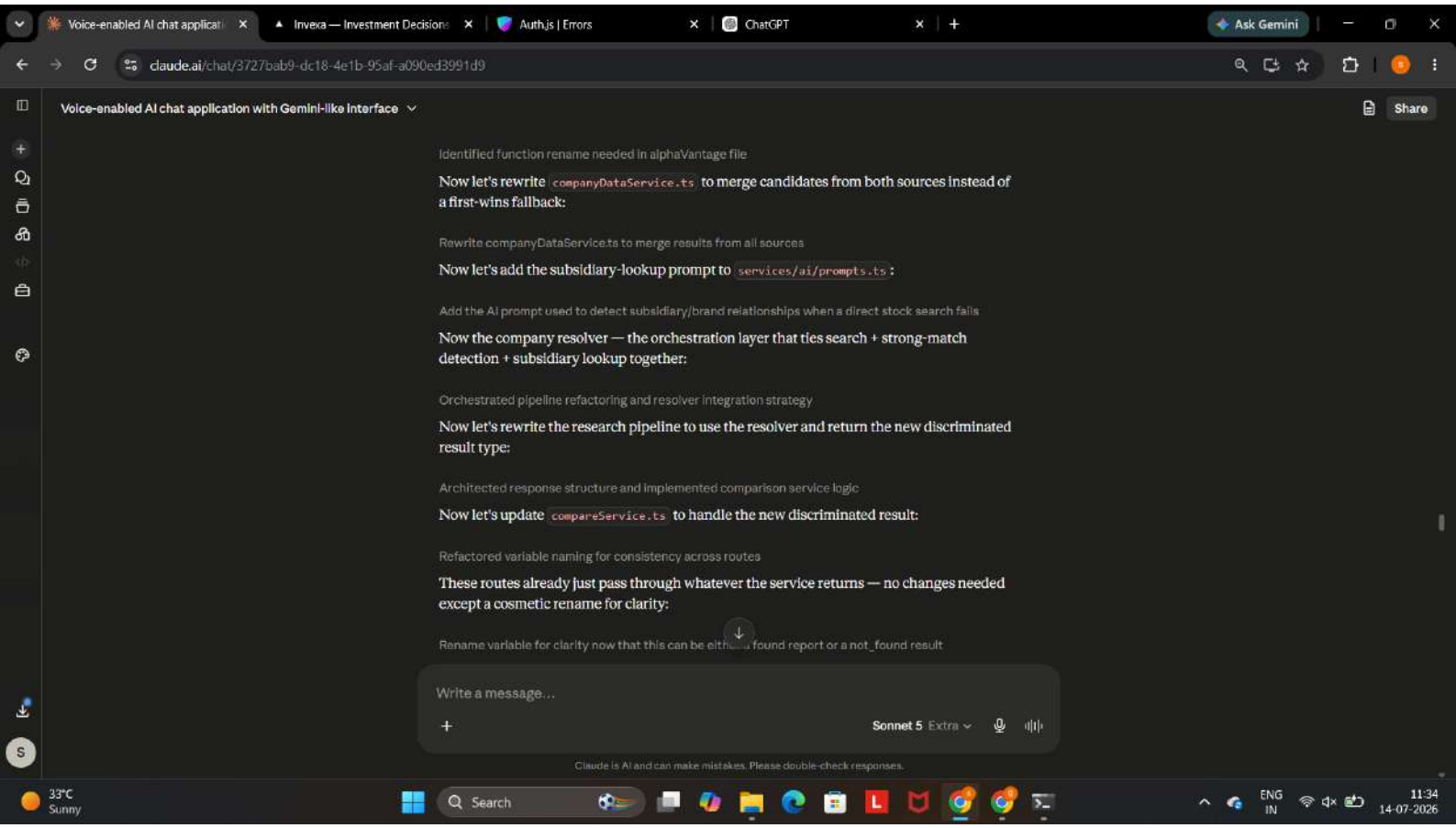
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Identified function rename needed in alphaVantage file
Now let's rewrite companyDataService.ts to merge candidates from both sources instead of a first-wins fallback:
Rewrite companyDataService.ts to merge results from all sources
Now let's add the subsidiary-lookup prompt to services/ai/prompts.ts:
Add the AI prompt used to detect subsidiary/brand relationships when a direct stock search fails
Now the company resolver — the orchestration layer that ties search + strong-match detection + subsidiary lookup together:
Orchestrated pipeline refactoring and resolver integration strategy
Now let's rewrite the research pipeline to use the resolver and return the new discriminated result type:
Architected response structure and implemented comparison service logic
Now let's update compareService.ts to handle the new discriminated result:
Refactored variable naming for consistency across routes
These routes already just pass through whatever the service returns — no changes needed except a cosmetic rename for clarity:
Rename variable for clarity now that this can be eith found report or a not_found result
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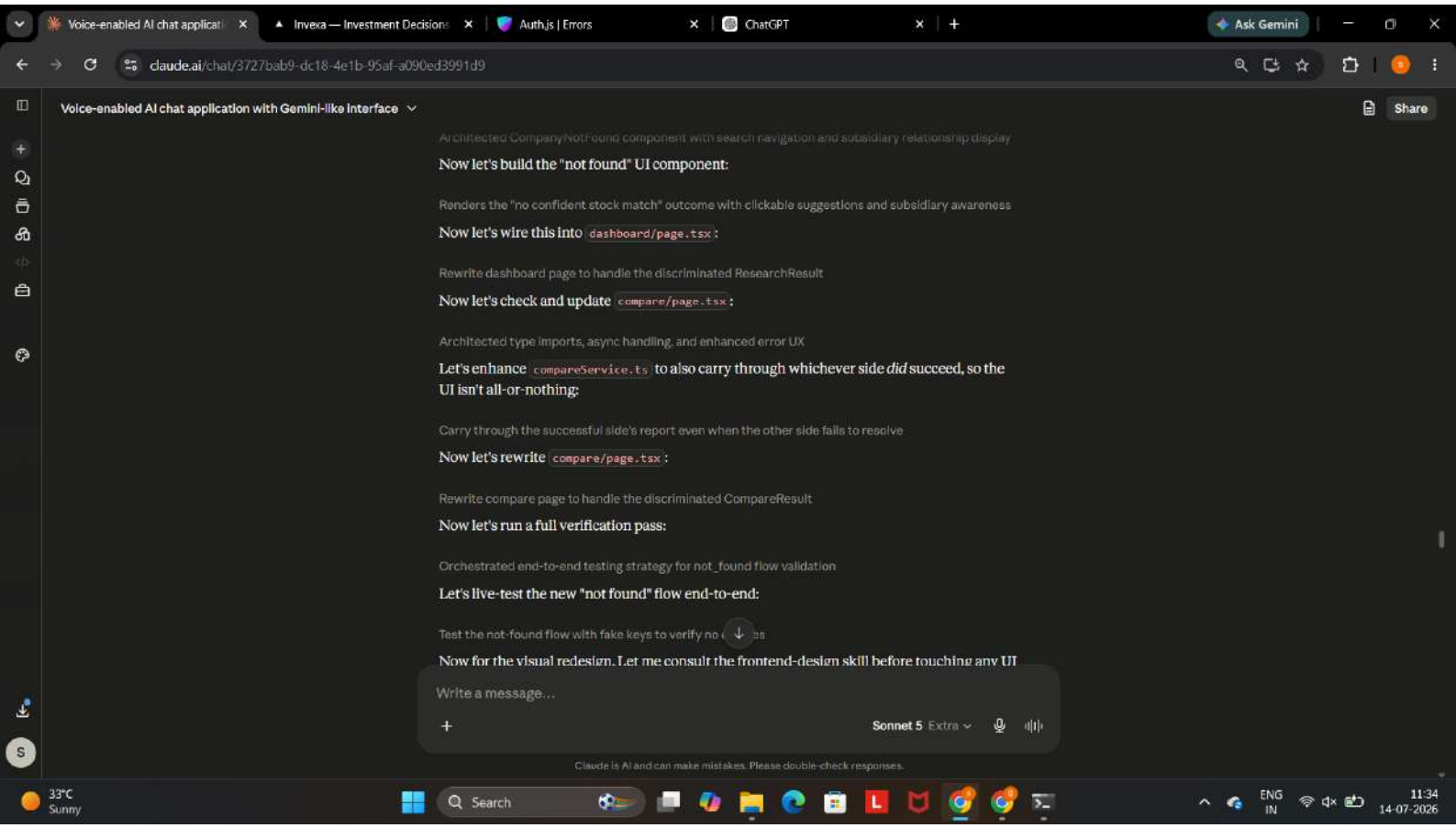
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Architected CompanyNotFound component with search navigation and subsidiary relationship display
Now let's build the "not found" UI component:
Renders the "no confident stock match" outcome with clickable suggestions and subsidiary awareness
Now let's wire this into dashboard/page.tsx:
Rewrite dashboard page to handle the discriminated ResearchResult
Now let's check and update compare/page.tsx:
Architected type imports, async handling, and enhanced error UX
Let's enhance compareService.ts to also carry through whichever side did succeed, so the UI isn't all-or-nothing:
Carry through the successful side's report even when the other side fails to resolve
Now let's rewrite compare/page.tsx:
Rewrite compare page to handle the discriminated CompareResult
Now let's run a full verification pass:
Orchestrated end-to-end testing strategy for not_found flow validation
Let's live-test the new "not found" flow end-to-end:
Test the not-found flow with fake keys to verify no
Now for the visual redesign. Let me consult the frontend-design skill before touching any UI
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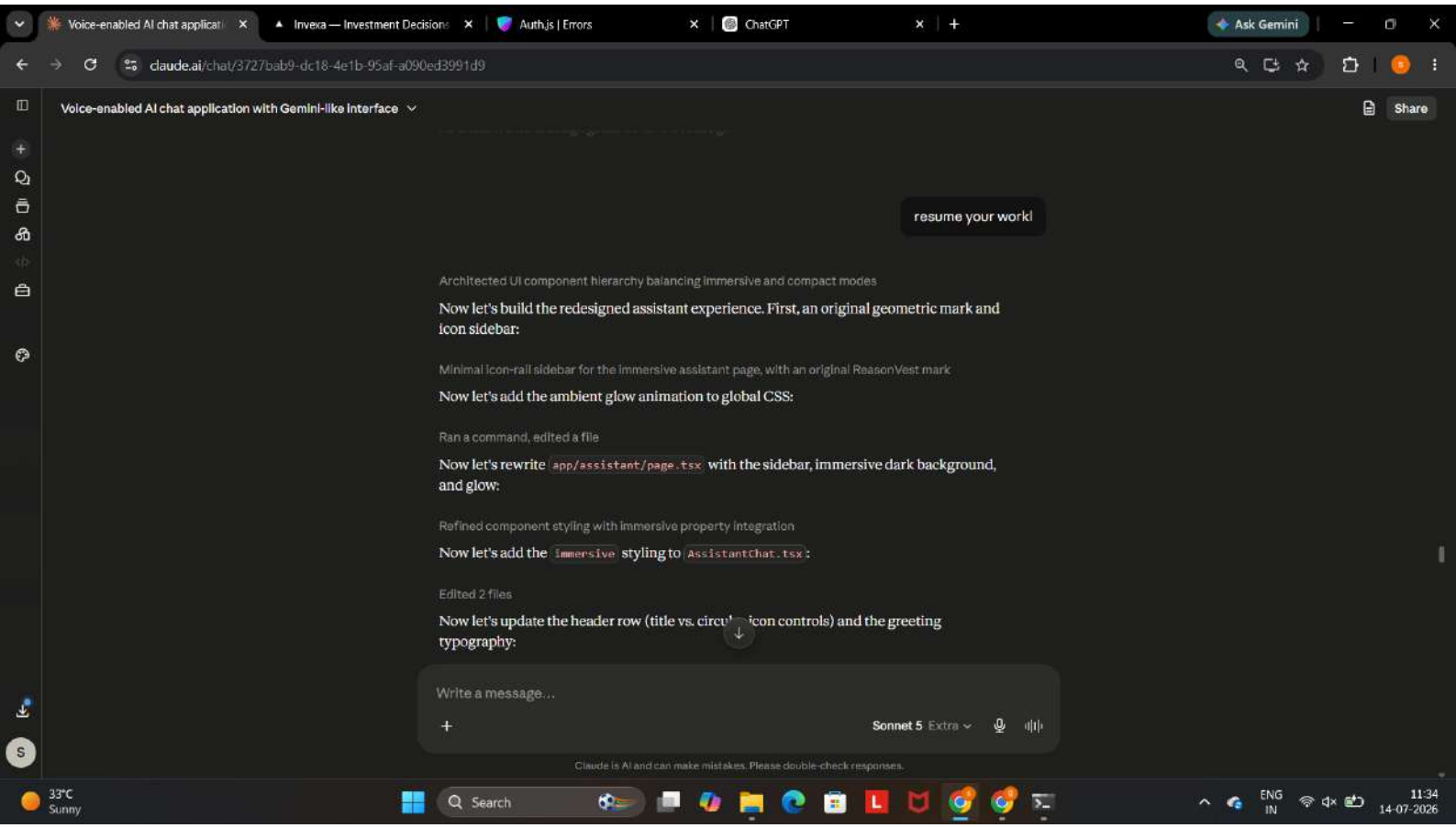
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Architected UI component hierarchy balancing immersive and compact modes
Now let's build the redesigned assistant experience. First, an original geometric mark and icon sidebar:
Minimal icon-rail sidebar for the immersive assistant page, with an original ReasonVest mark
Now let's add the ambient glow animation to global CSS:
Ran a command, edited a file
Now let's rewrite app/assistant/page.tsx with the sidebar, immersive dark background, and glow:
Refined component styling with immersive property integration
Now let's add the immersive styling to AssistantChat.tsx:
Edited 2 files
Now let's update the header row (title vs. circ icon controls) and the greeting typography:
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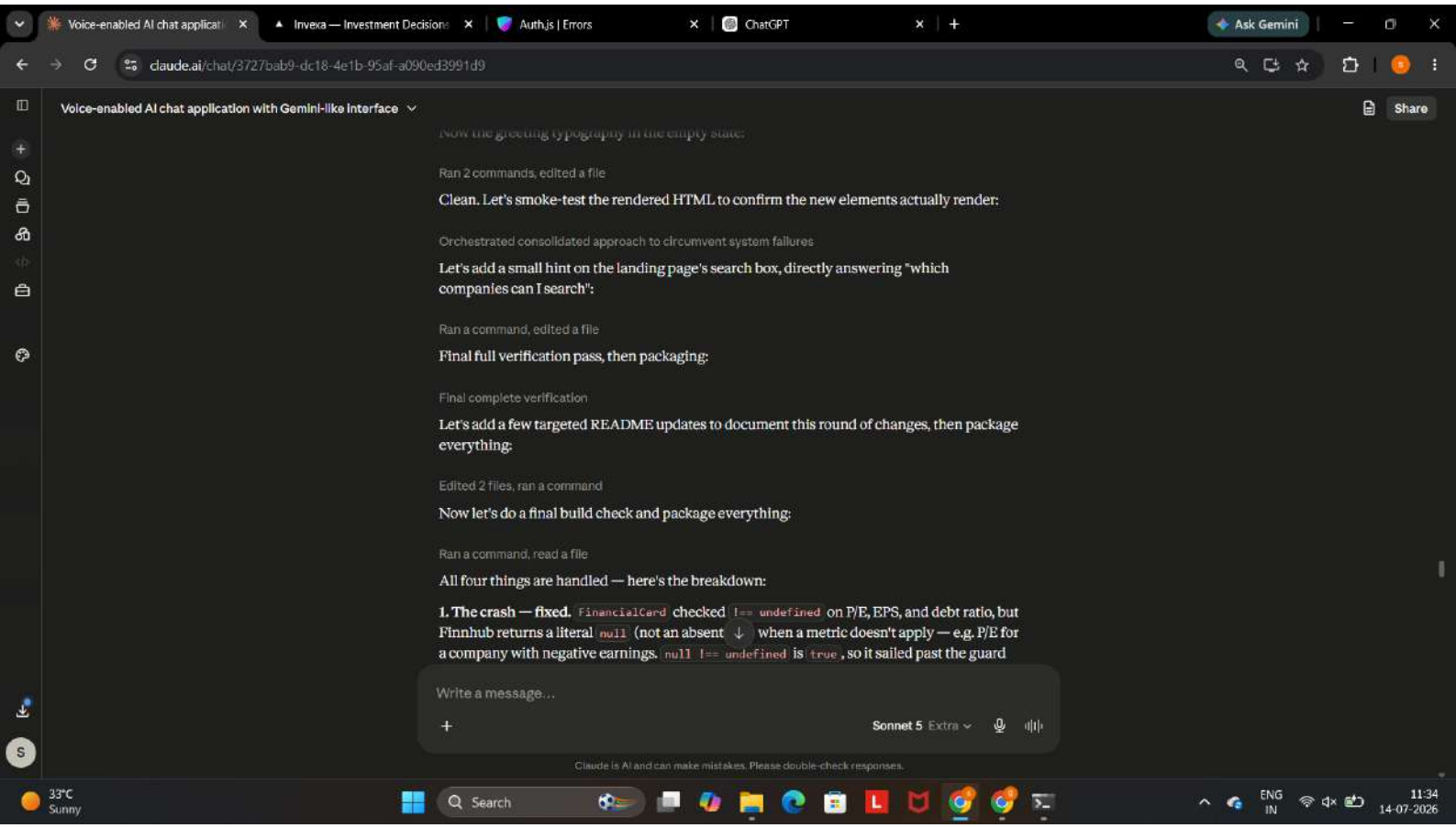
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Now the greeting typography in the empty state:
Ran 2 commands, edited a file
Clean. Let's smoke-test the rendered HTML to confirm the new elements actually render:
Orchestrated consolidated approach to circumvent system failures
Let's add a small hint on the landing page's search box, directly answering "which companies can I search":
Ran a command, edited a file
Final full verification pass, then packaging:
Final complete verification
Let's add a few targeted README updates to document this round of changes, then package everything:
Edited 2 files, ran a command
Now let's do a final build check and package everything:
Ran a command, read a file
All four things are handled — here's the breakdown:
1. The crash — fixed. FinancialCard checked !== undefined on P/E, EPS, and debt ratio, but Finnhub returns a literal null (not an absent ↓ when a metric doesn't apply — e.g. P/E for a company with negative earnings. null !== undefined is true, so it sailed past the guard
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ticker, or the name genuinely contains the query) before using it — "bitcoin" no longer silently analyzes whatever a fuzzy search returned first. When nothing confident is found, you get real alternative tickers from the search results, plus — when applicable — the AI checks if the term is a subsidiary/brand of a public company (Instagram → Meta) and that guess gets **independently re-verified** against real search data before it's ever shown, so a hallucinated ticker never reaches you. Same handling now flows through Compare too.

4. Assistant redesign — done. `/assistant` now has an icon-rail sidebar (every icon is real navigation, not decoration), a large soft-weight greeting, circular icon controls instead of labeled pills, and a slow ambient glow behind the composer. I kept it visually part of ReasonVest AI rather than a literal clone — original mark, your existing blue/cyan palette — since copying Gemini's actual logo/branding isn't something I'll do, but the *structure* you pointed at (minimal chrome, icon rail, centered glow-lit input) is fully there.

Everything above is verified — typecheck, lint, and build all clean, plus I live-tested the crash fix, the not-found flow, and the redesigned page against a running server.

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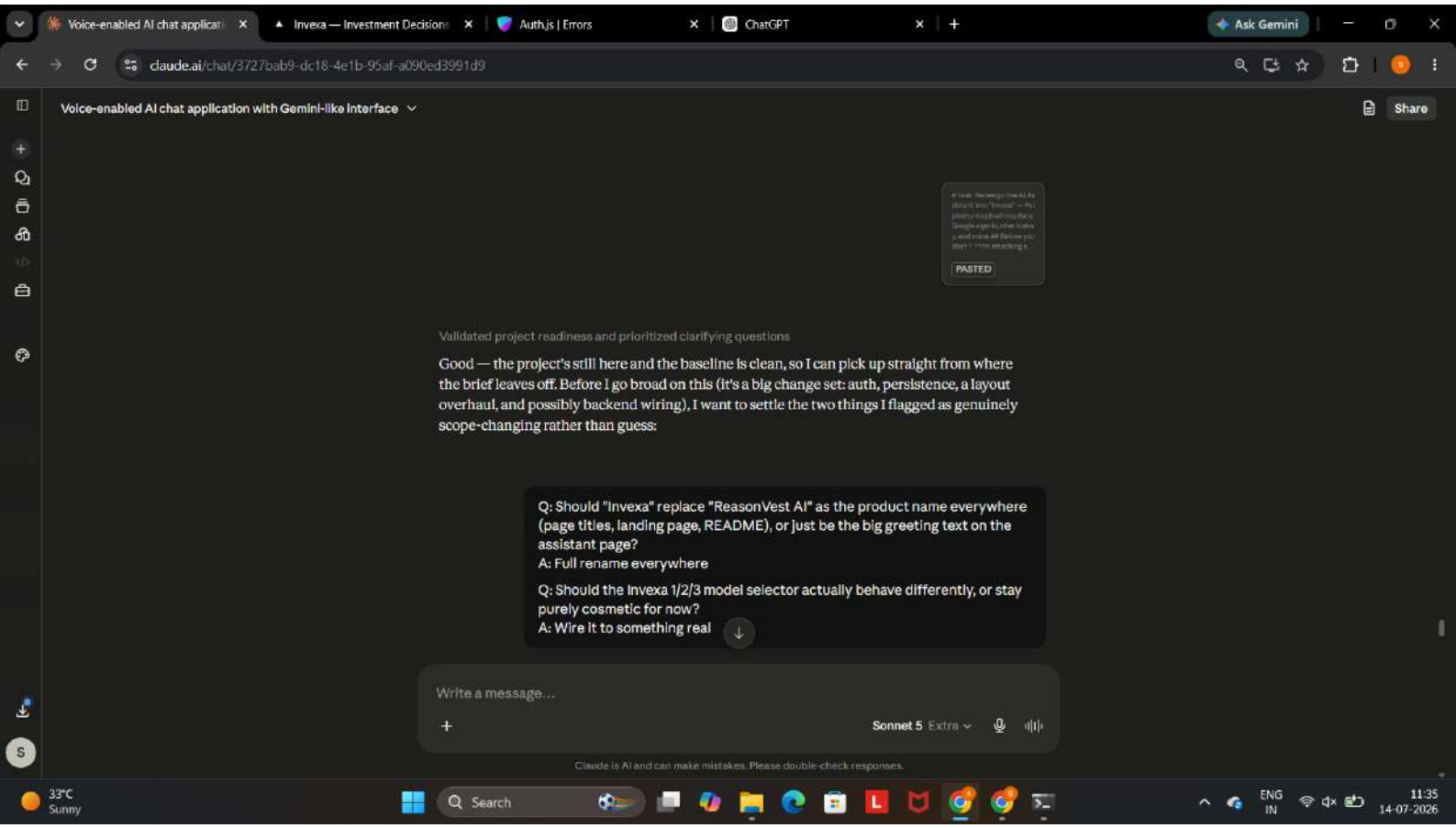
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PASTED
Validated project readiness and prioritized clarifying questions
Good — the project's still here and the baseline is clean, so I can pick up straight from where the brief leaves off. Before I go broad on this (it's a big change set: auth, persistence, a layout overhaul, and possibly backend wiring), I want to settle the two things I flagged as genuinely scope-changing rather than guess:
Q: Should "Invexa" replace "ReasonVest AI" as the product name everywhere (page titles, landing page, README), or just be the big greeting text on the assistant page?
A: Full rename everywhere
Q: Should the Invexa 1/2/3 model selector actually behave differently, or stay purely cosmetic for now?
A: Wire it to something real
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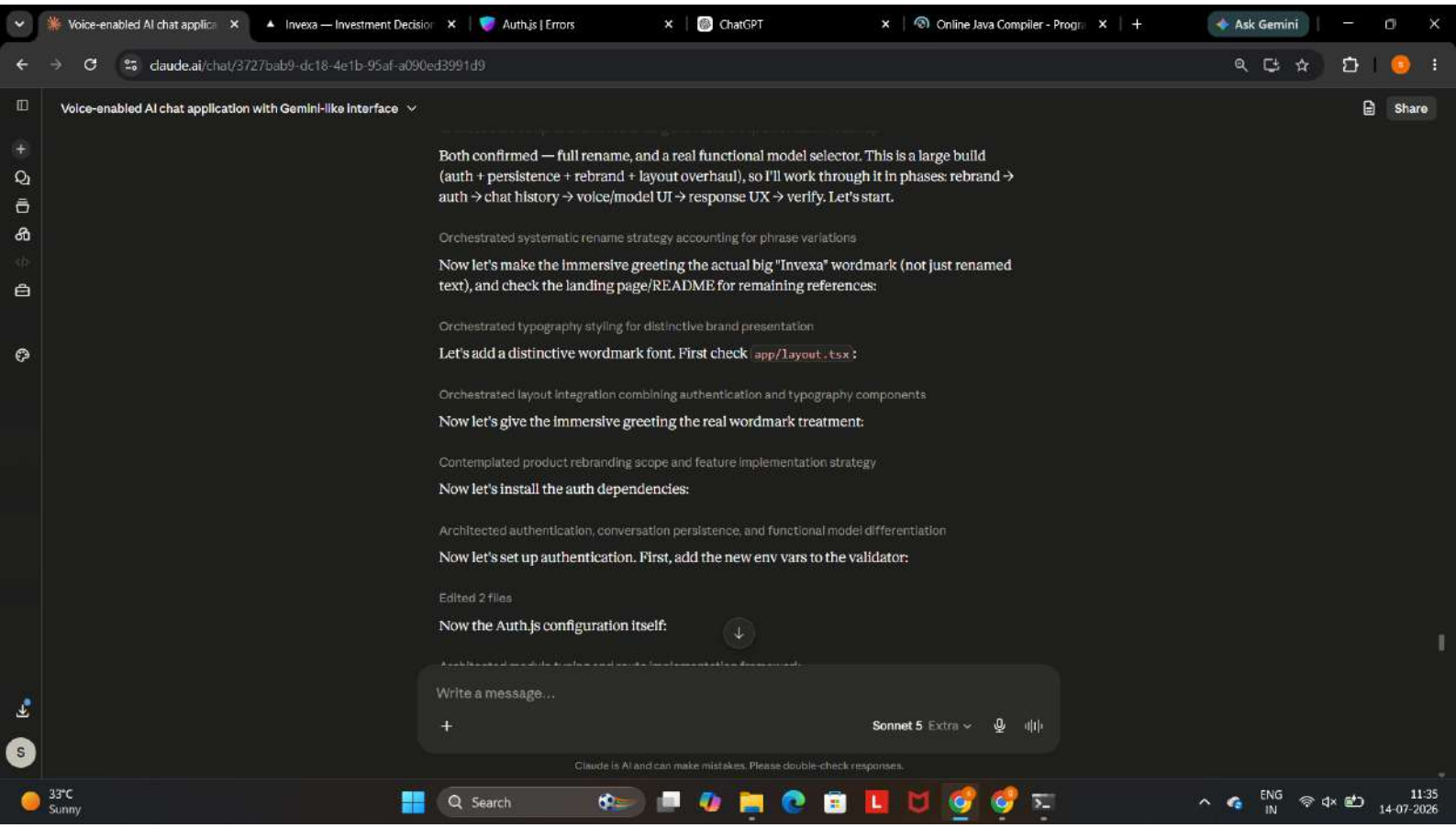
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Both confirmed — full rename, and a real functional model selector. This is a large build (auth + persistence + rebrand + layout overhaul), so I'll work through it in phases: rebrand → auth → chat history → voice/model UI → response UX → verify. Let's start.
Orchestrated systematic rename strategy accounting for phrase variations
Now let's make the immersive greeting the actual big "Invexa" wordmark (not just renamed text), and check the landing page/README for remaining references:
Orchestrated typography styling for distinctive brand presentation
Let's add a distinctive wordmark font. First check app/layout.tsx:
Orchestrated layout integration combining authentication and typography components
Now let's give the immersive greeting the real wordmark treatment:
Contemplated product rebranding scope and feature implementation strategy
Now let's install the auth dependencies:
Architected authentication, conversation persistence, and functional model differentiation
Now let's set up authentication. First, add the new env vars to the validator:
Edited 2 files
Now the Auth.js configuration itself:
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Auth core compiles clean. Now the conversation repository for chat history:

Architected message sanitization strategy for database storage optimization

Let's strip attachment binary data before persisting to MongoDB — storing raw base64 blobs per message would bloat the database and risk hitting document size limits:

Edited 3 files

Now let's add the request schemas for conversation routes:

Add schemas for the chat history API routes

Now the API routes for conversations:

Architected backend threading and UI integration for model selector

Now let's wire the model selector to something real. First, the AI layer plumbing:

Add forceProvider to AIRequest so a caller can pin a specific provider instead of the default fallback order

Now update `providers.ts` to honor `forceProvider`:

Architected end-to-end model selection system with provider routing

Now let's define the model presets (this is where "cosmetic" becomes "real"):

Edited a file, ran a command

Let me add the import and the model presets ↓ ly at the top and near the relevant section:

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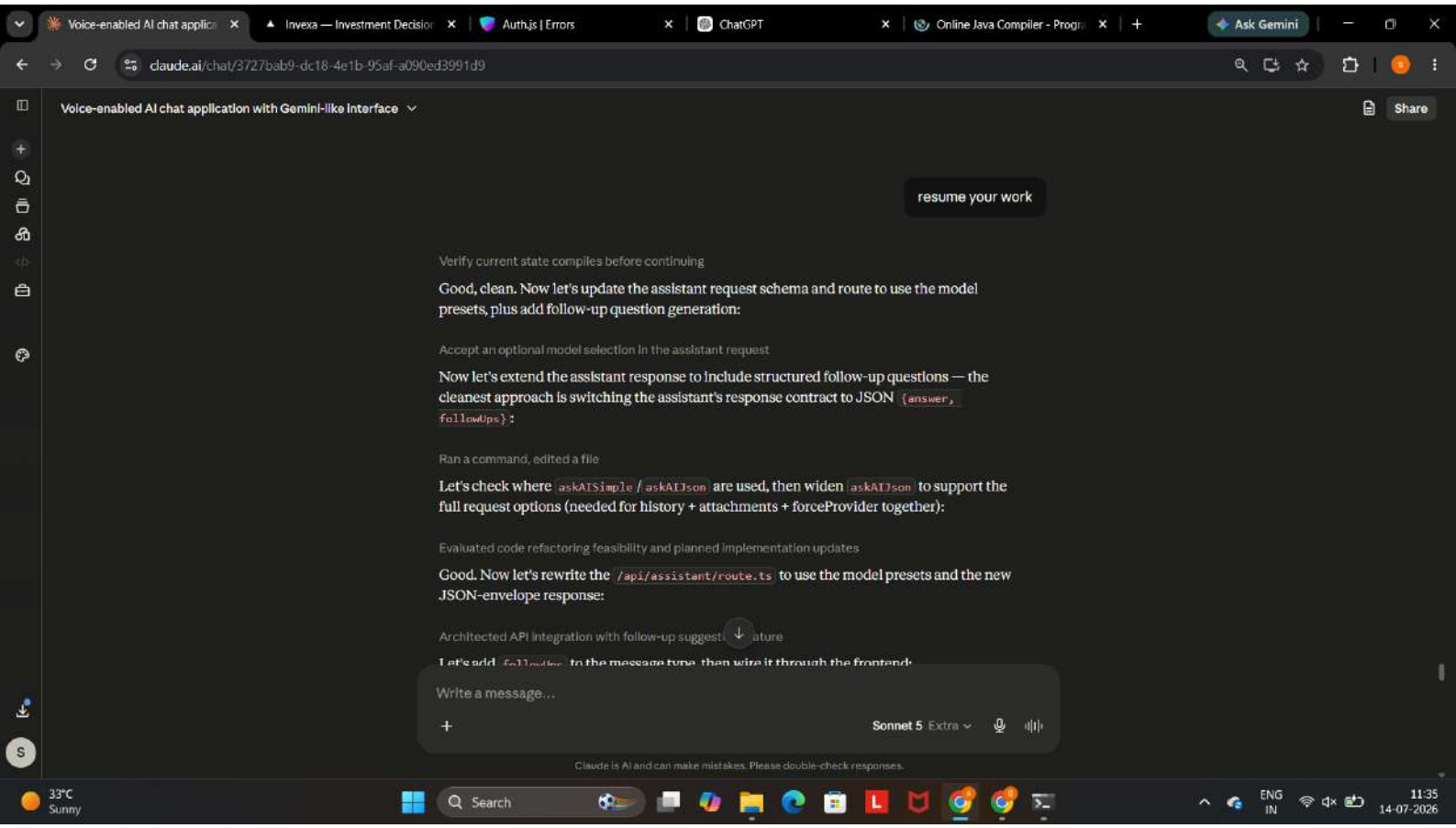
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Verify current state compiles before continuing
Good, clean. Now let's update the assistant request schema and route to use the model presets, plus add follow-up question generation:
Accept an optional model selection in the assistant request
Now let's extend the assistant response to include structured follow-up questions — the cleanest approach is switching the assistant's response contract to JSON {answer, followUps}:
Ran a command, edited a file
Let's check where askAISimple / askAIJson are used, then widen askAIJson to support the full request options (needed for history + attachments + forceProvider together):
Evaluated code refactoring feasibility and planned implementation updates
Good. Now let's rewrite the /api/assistant/route.ts to use the model presets and the new JSON-envelope response:
Architected API integration with follow-up suggesti ature
Let's add followUps to the message type, then wire it through the frontend:
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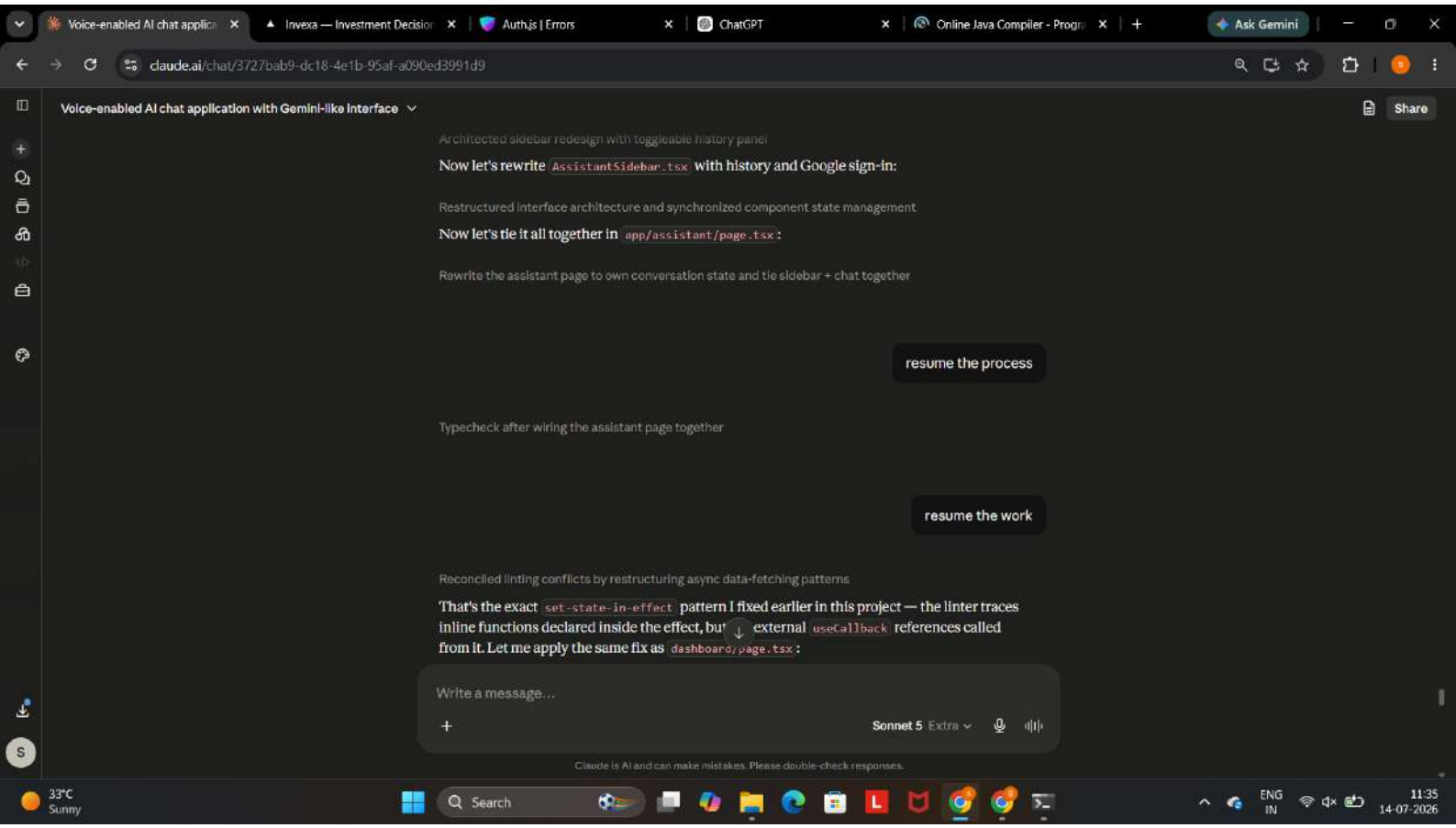
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Architected sidebar redesign with toggleable history panel
Now let's rewrite AssistantSidebar.tsx with history and Google sign-in:
Restructured interface architecture and synchronized component state management
Now let's tie it all together in app/assistant/page.tsx:
Rewrite the assistant page to own conversation state and tie sidebar + chat together
resume the process
Typecheck after wiring the assistant page together
resume the work
Reconciled linting conflicts by restructuring async data-fetching patterns
That's the exact set-state-in-effect pattern I fixed earlier in this project — the linter traces inline functions declared inside the effect, but external useCallback references called from it. Let me apply the same fix as dashboard/page.tsx:
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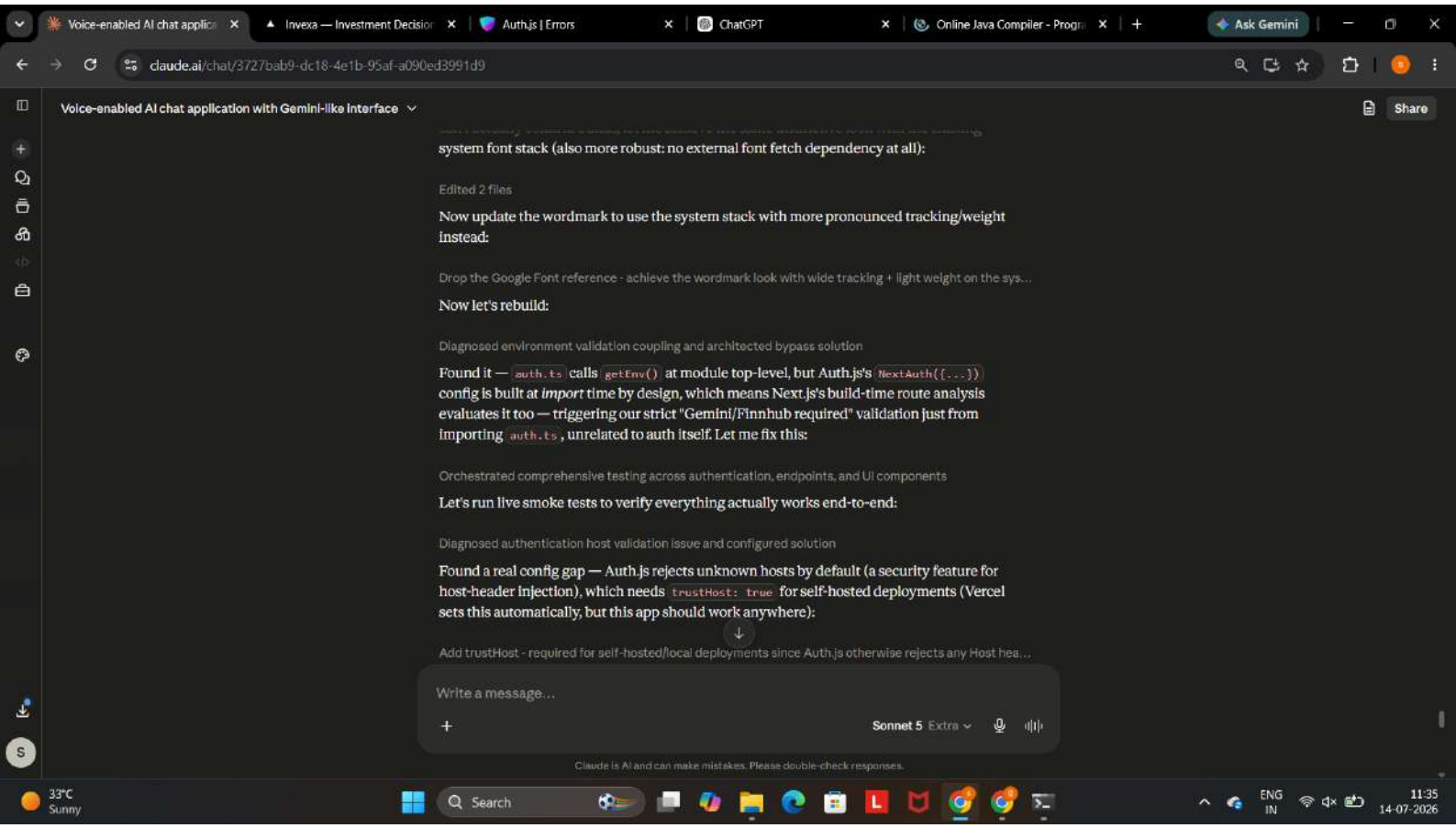
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system font stack (also more robust: no external font fetch dependency at all):
Edited 2 files
Now update the wordmark to use the system stack with more pronounced tracking/weight instead:
Drop the Google Font reference - achieve the wordmark look with wide tracking + light weight on the sys...
Now let's rebuild:
Diagnosed environment validation coupling and architected bypass solution
Found it — auth.ts calls getEnv() at module top-level, but Auth.js's NextAuth({...}) config is built at import time by design, which means Next.js's build-time route analysis evaluates it too — triggering our strict "Gemini/Finnhub required" validation just from importing auth.ts, unrelated to auth itself. Let me fix this:
Orchestrated comprehensive testing across authentication, endpoints, and UI components
Let's run live smoke tests to verify everything actually works end-to-end:
Diagnosed authentication host validation issue and configured solution
Found a real config gap — Auth.js rejects unknown hosts by default (a security feature for host-header injection), which needs trustHost: true for self-hosted deployments (Vercel sets this automatically, but this app should work anywhere):
Add trustHost - required for self-hosted/local deployments since Auth.js otherwise rejects any Host hea...
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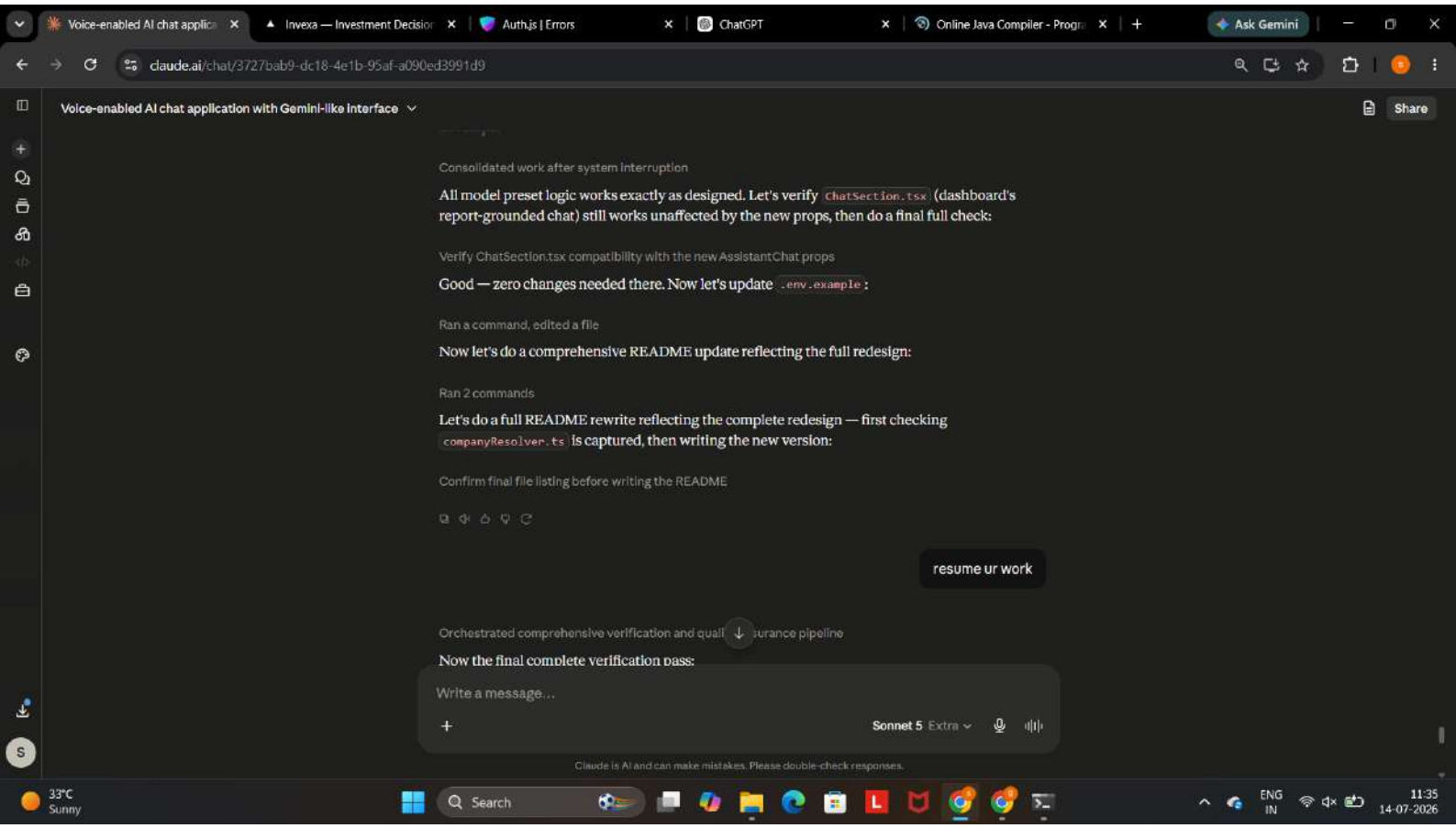
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Consolidated work after system interruption
All model preset logic works exactly as designed. Let's verify ChatSection.tsx (dashboard's report-grounded chat) still works unaffected by the new props, then do a final full check:
Verify ChatSection.tsx compatibility with the new AssistantChat props
Good — zero changes needed there. Now let's update .env.example:
Ran a command, edited a file
Now let's do a comprehensive README update reflecting the full redesign:
Ran 2 commands
Let's do a full README rewrite reflecting the complete redesign — first checking companyResolver.ts is captured, then writing the new version:
Confirm final file listing before writing the README
resume ur work
Orchestrated comprehensive verification and quality assurance pipeline
Now the final complete verification pass:
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big, thin-weight treatment on the system font, not a Google Fonts fetch — that actually broke the build in this sandbox since it needs network access I couldn't verify, so I built the same look without the external dependency).

Google sign-in. Auth.js v5, JWT sessions (no DB adapter — unnecessary for one provider), fully optional: the app works exactly as before when it's not configured.

Chat history. MongoDB-backed, scoped per signed-in user, listed in a sidebar flyout. Attachment binary data is stripped before saving — full base64 blobs in every message would bloat the collection fast.

Voice, two ways. The composer's mic (as before) plus a genuinely separate **voice mode** — the circular icon opens a hands-free loop that listens, sends, speaks the reply, and listens again on its own.

A model selector that's real, not decorative — you picked "wire it to something real," so: Invexa 1 is the default chain, Invexa 2 pins to OpenRouter specifically, Invexa 3 keeps the chain but asks for a livelier tone. All three go through the same fallback/retry/error-handling pipeline.

Follow-ups — every answer ships with contextual next-question chips, generated in the same JSON-mode call as the answer itself.

A few real bugs surfaced and got fixed along the way (all documented in the README, same honesty-over-hiding pattern as the rest of this project): `auth.ts` was triggering full env validation just from being imported, breaking builds unrelated to auth entirely; Auth.js needed `trustHost` for anything not on Vercel; the MongoDB adapter needed a v6 downgrade; and two real React Compiler lint c hes (an effect calling an opaque external function, and a temporal-dead-zone reference, ot fixed rather than suppressed.

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